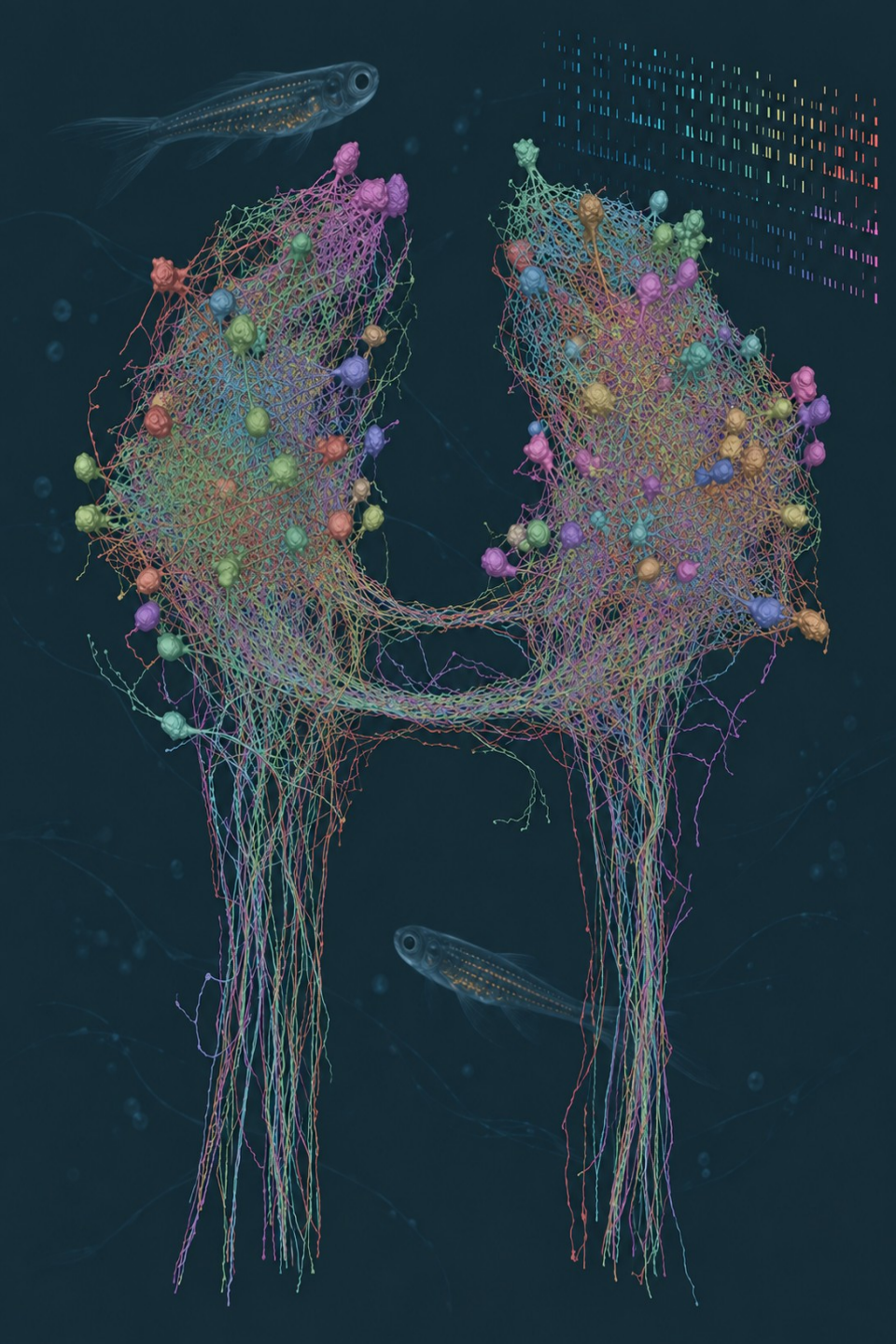




Connectome-constrained spiking network models of functional activity

Rory Bedford

Bernstein Satellite Workshop
September 28th 2026



Friedrich Lab



Zenke Lab

Talk Outline

Talk Outline

Part 1: Elucidating the implementation of a spatial cognitive map using connectome-constrained neural network models

Talk Outline

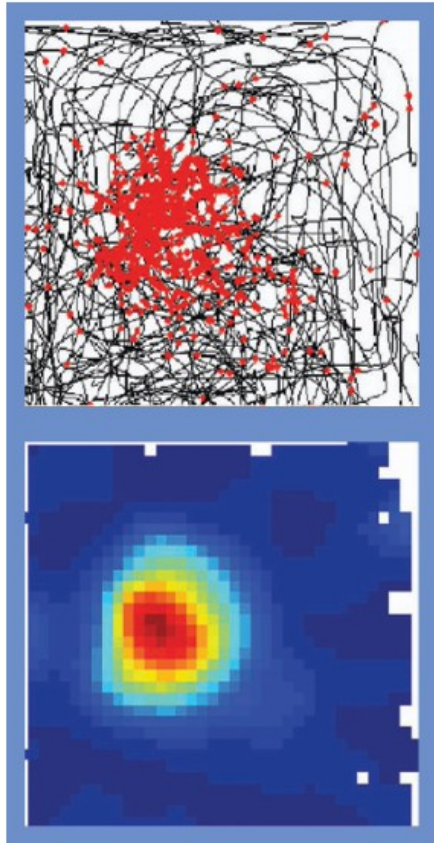
Part 1: Elucidating the implementation of a spatial cognitive map using connectome-constrained neural network models

Part 2: Robustness of circuit fits to realistic connectome degradations in a teacher-student learning paradigm

The Building Blocks of Spatial Cognitive Maps

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Place Cells

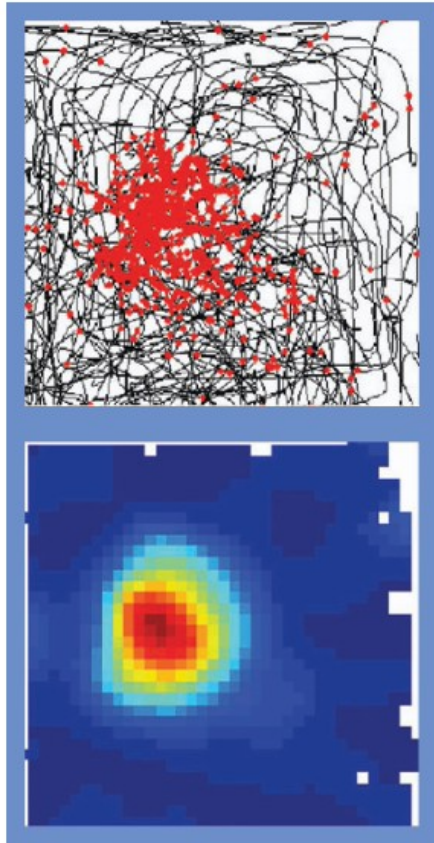


O'Keefe, J., & Dostrovsky, J. (1971)

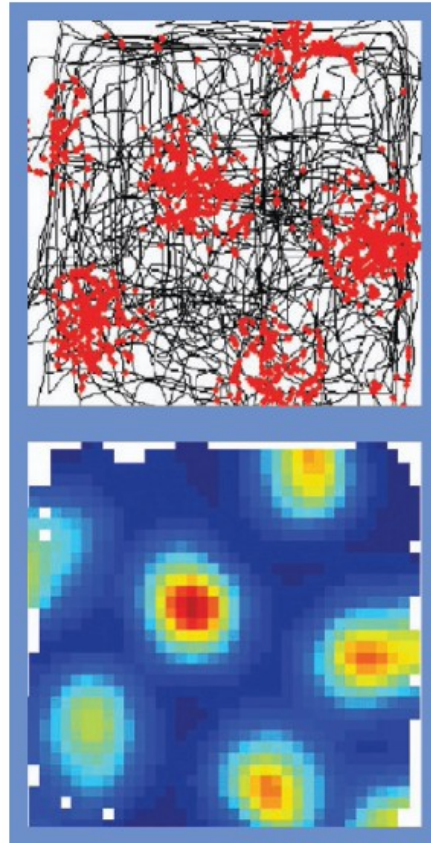
Figure adapted from Moser, M.-B., Rowland, D. C., & Moser, E. I. (2015)

The Building Blocks of Spatial Cognitive Maps

Place Cells

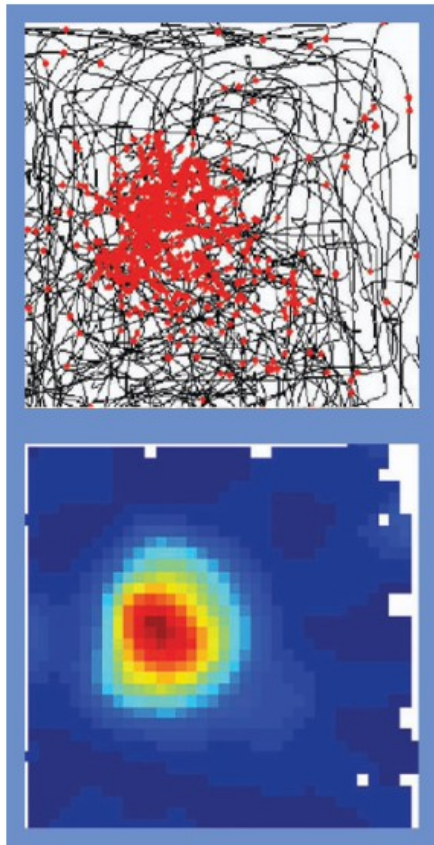


Grid Cells

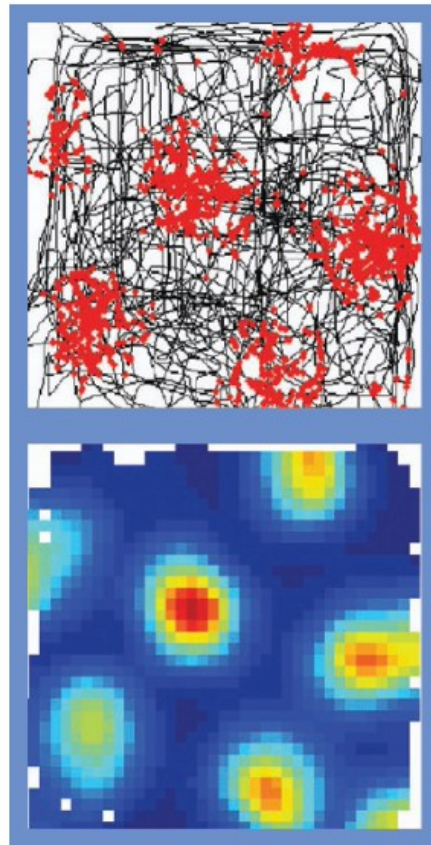


The Building Blocks of Spatial Cognitive Maps

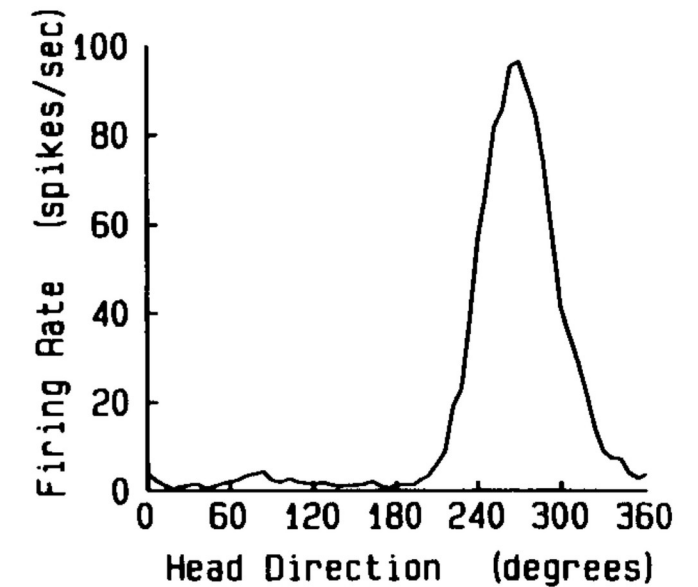
Place Cells



Grid Cells

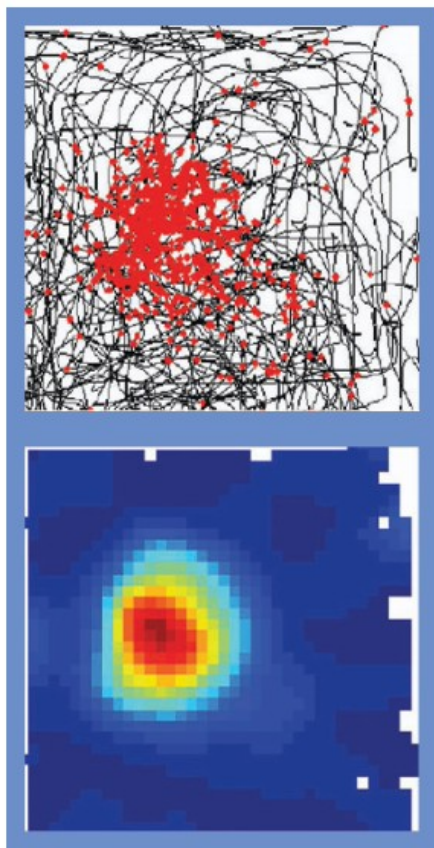


Head Direction Cells

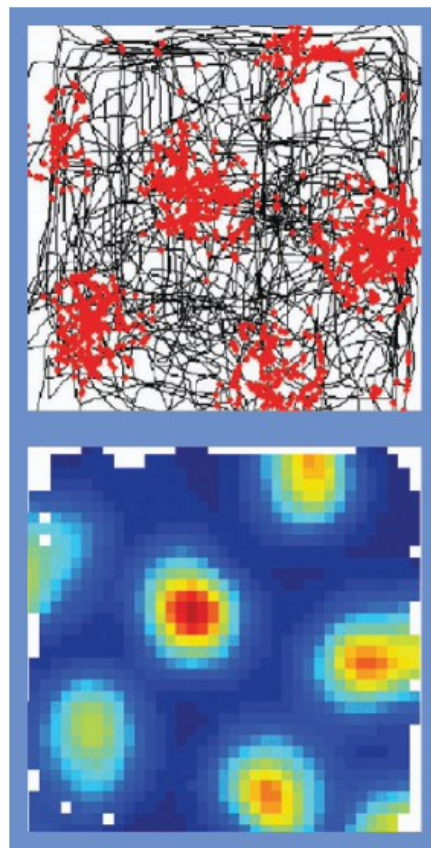


The Building Blocks of Spatial Cognitive Maps

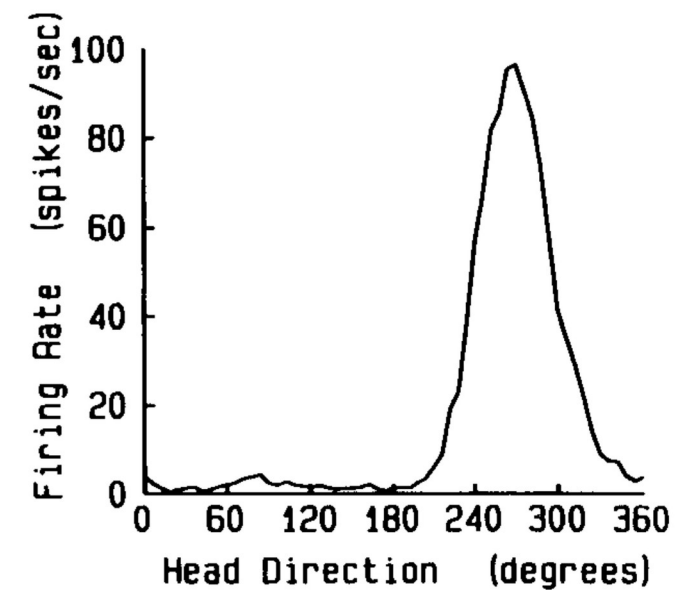
Place Cells



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Head Direction Cells

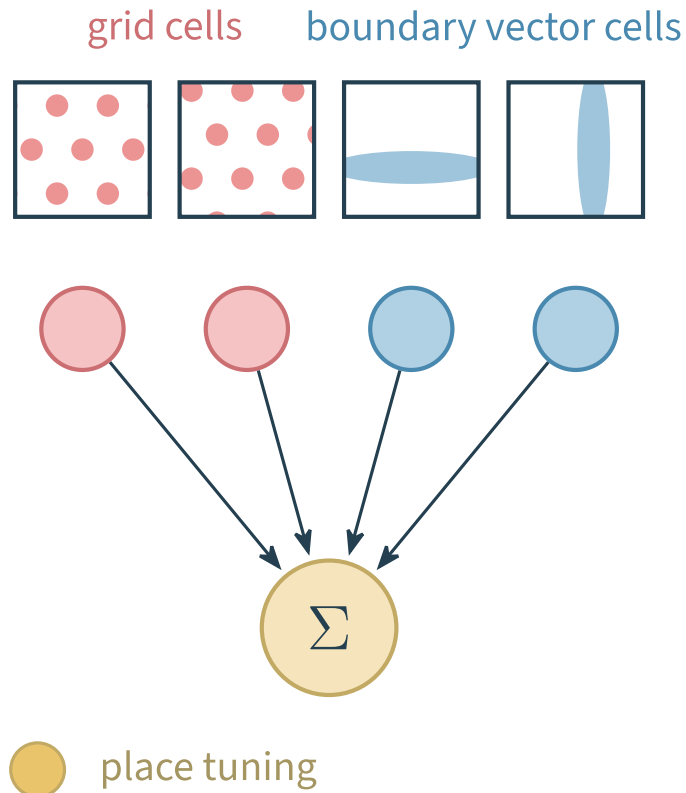


What Circuit Mechanisms Implement the Place Cell Map?

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Feedforward

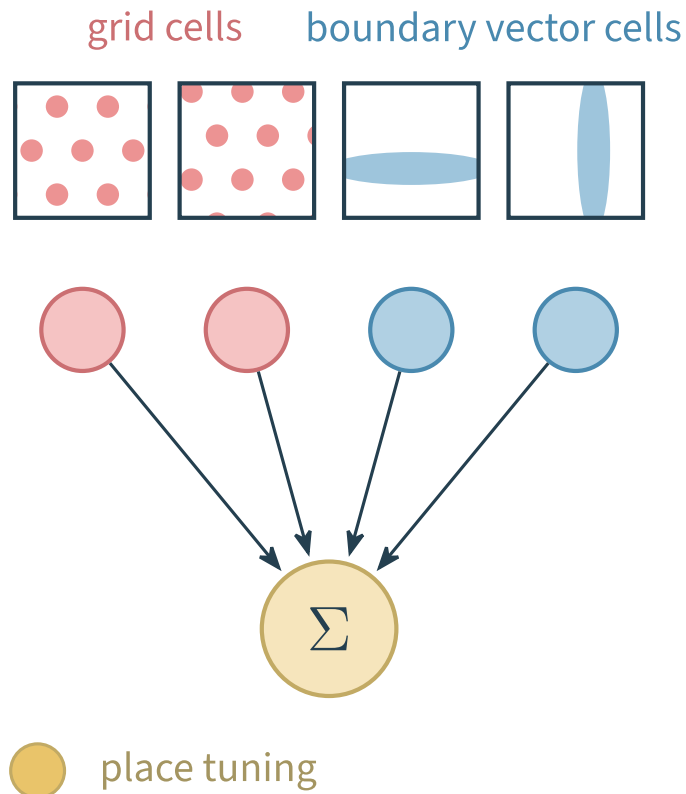
inherited by summing
spatial inputs



What Circuit Mechanisms Implement the Place Cell Map?

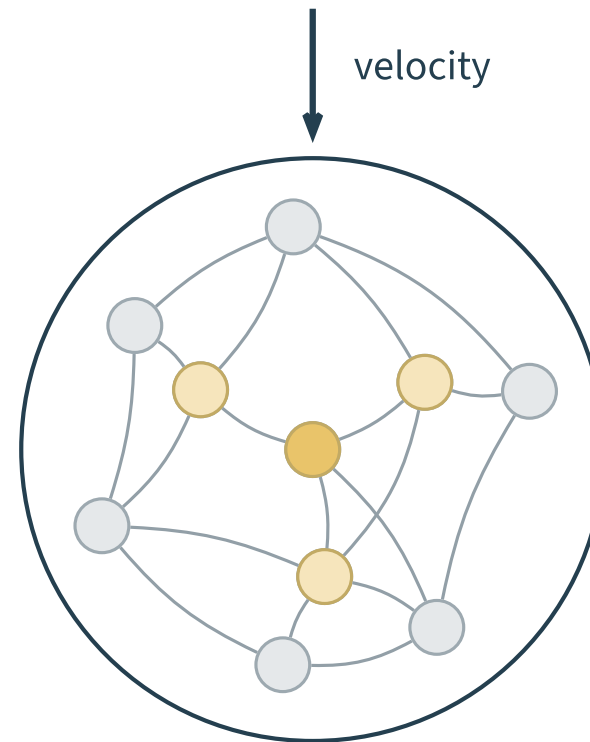
Feedforward

inherited by summing
spatial inputs



Recurrent

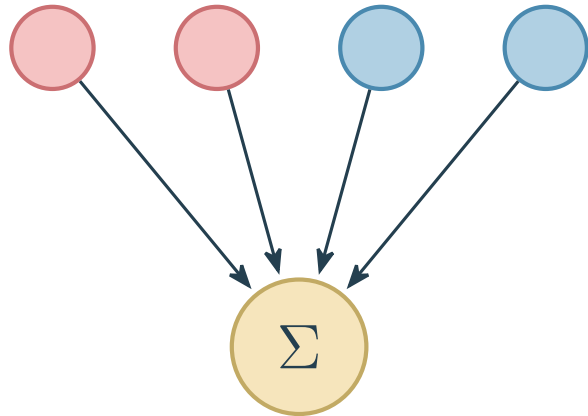
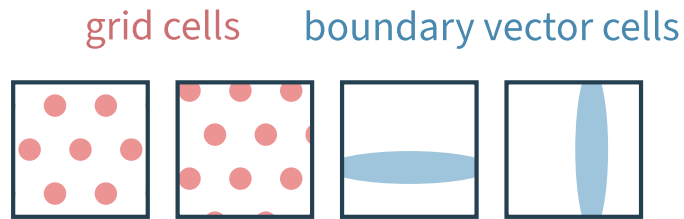
maintained by
attractor dynamics



What Circuit Mechanisms Implement the Place Cell Map?

Feedforward

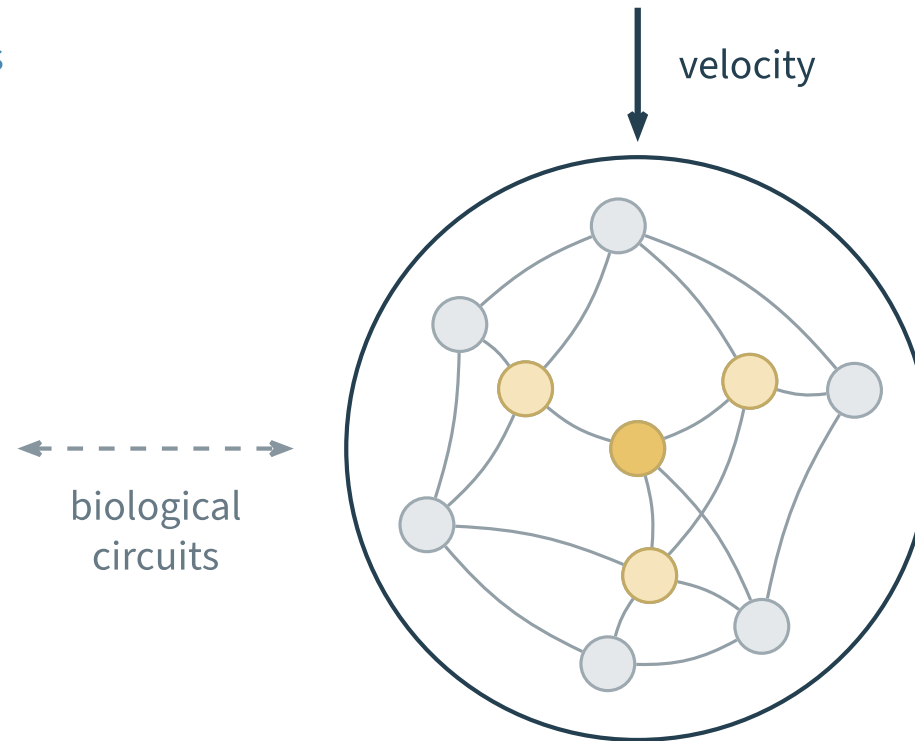
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● place tuning

Recurrent

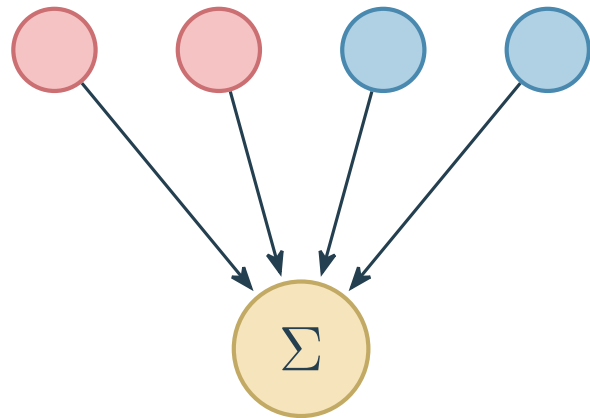
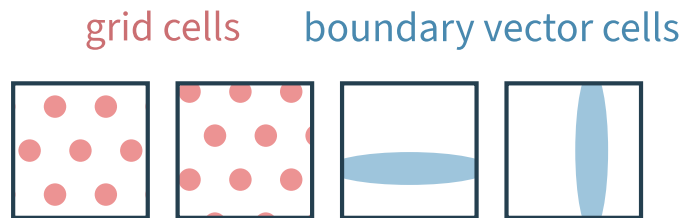
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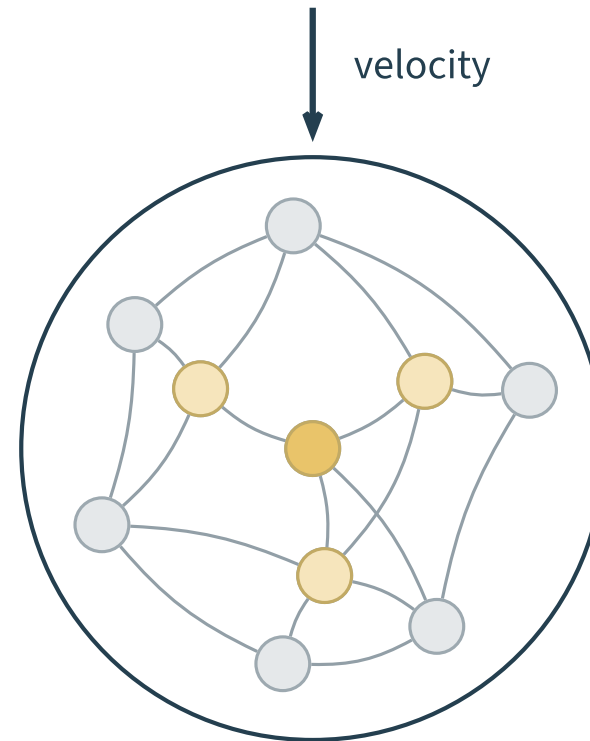
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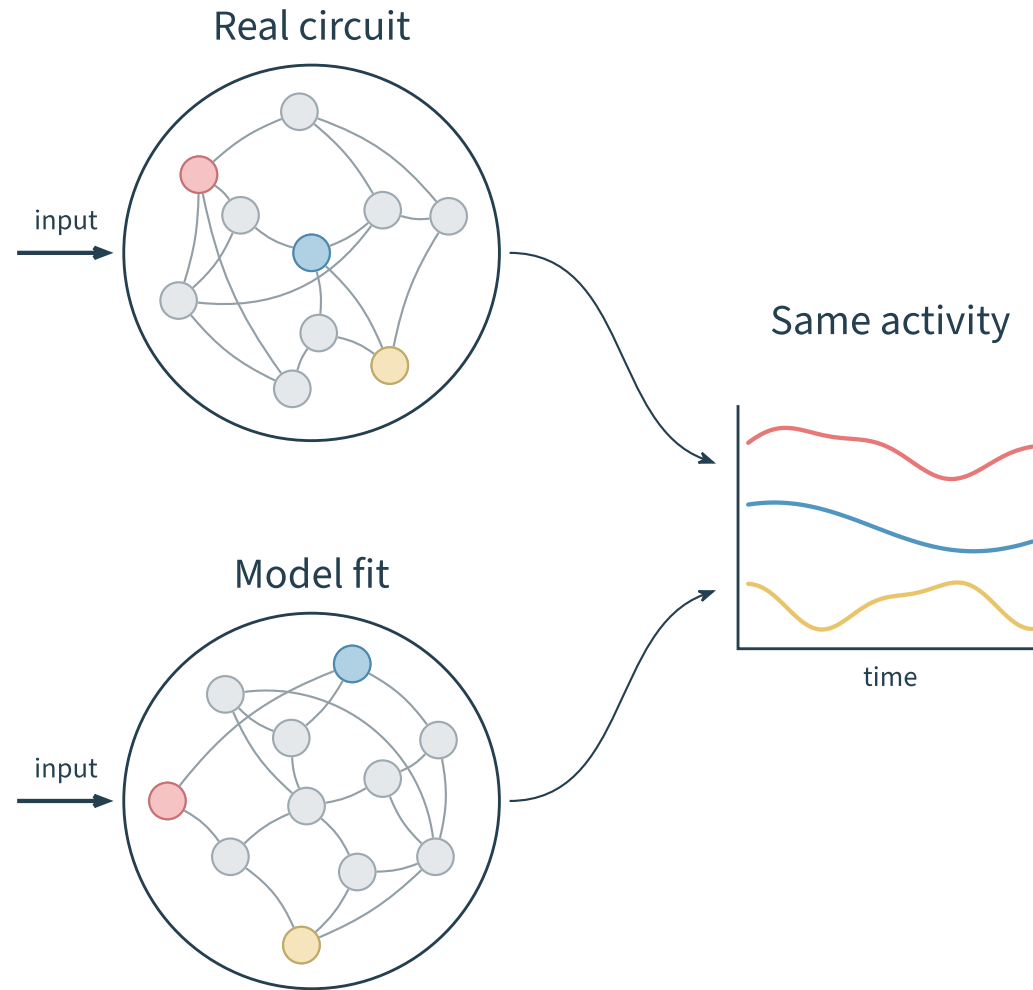
← biological
circuits →

Goal:

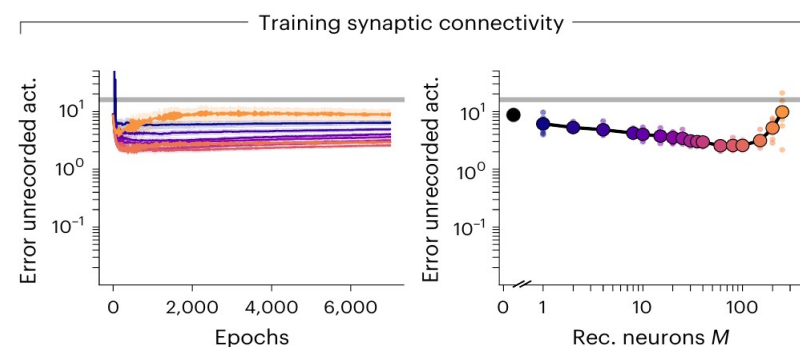
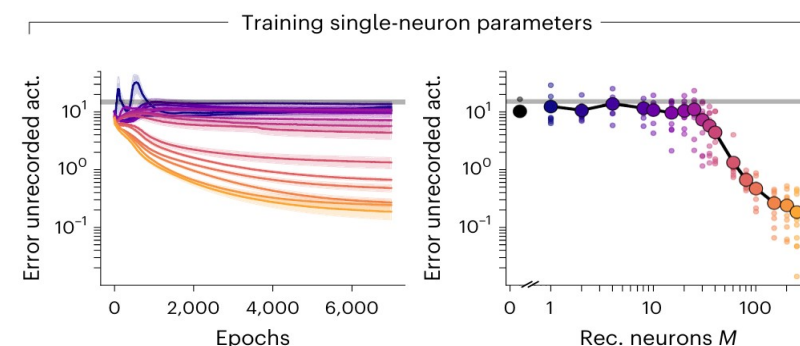
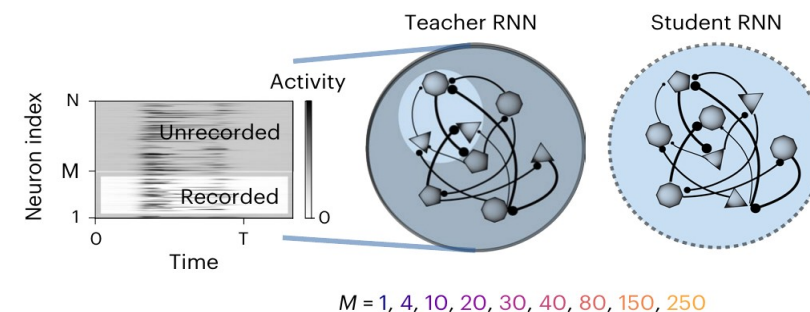
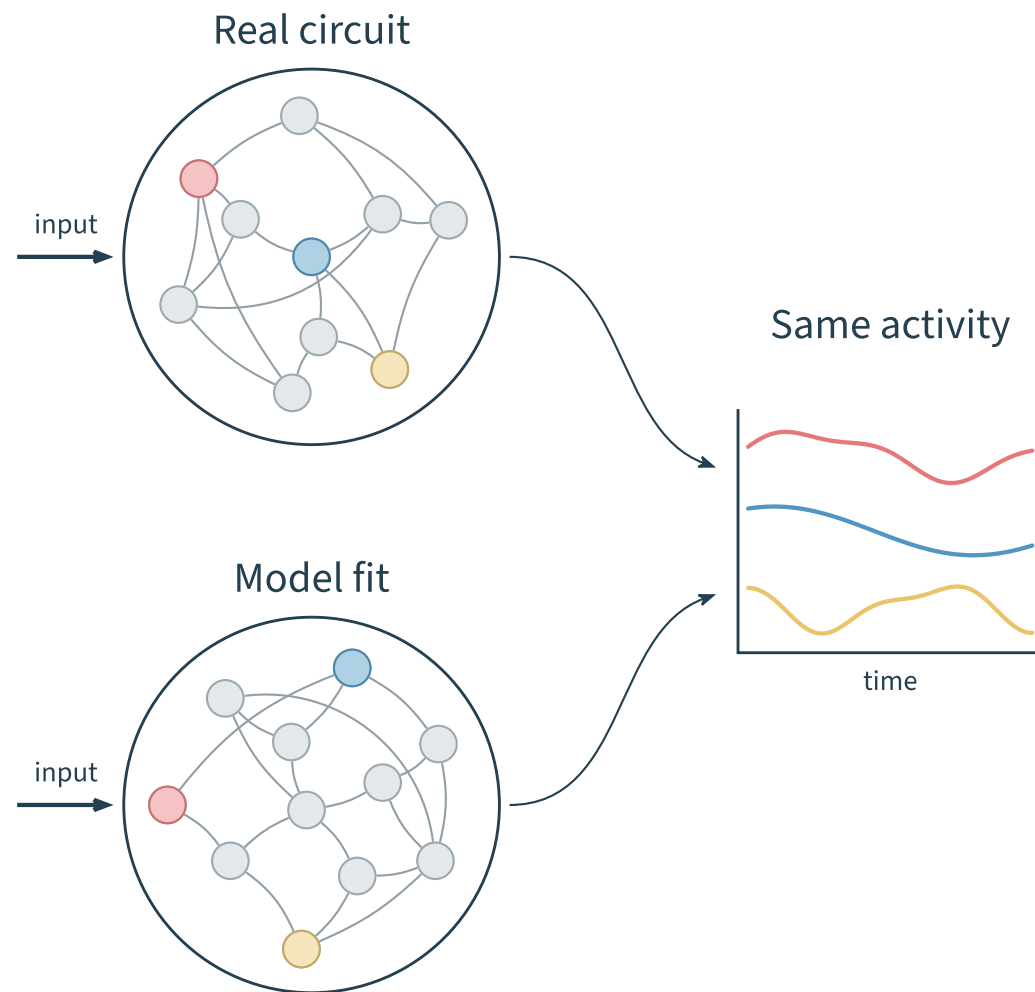
Can we fit network models that
allow us to disentangle the
mechanisms instantiating place
cells in real biological circuits?

Activity Alone Doesn't Constrain the Circuit, But Connectivity Constraints Help

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Activity Alone Doesn't Constrain the Circuit, But Connectivity Constraints Help

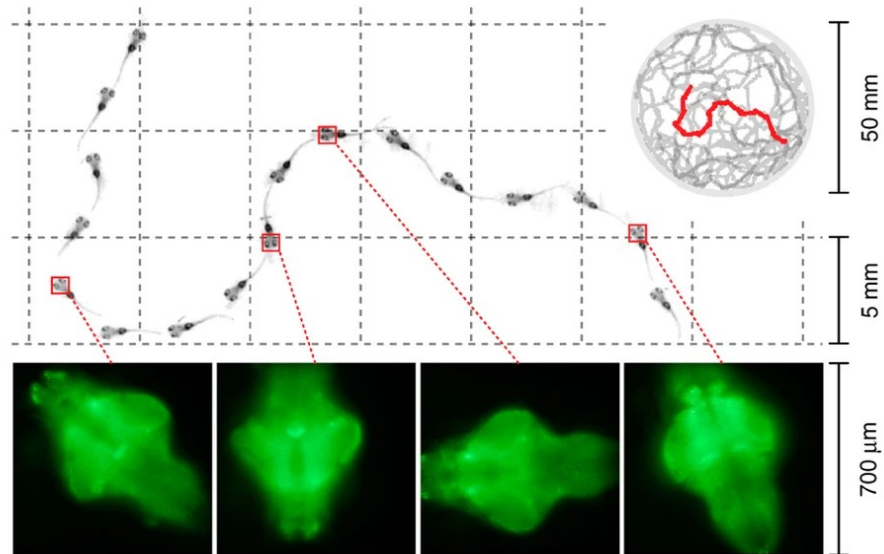


Place Cells in Larval Zebrafish (RoLi Lab, Tübingen)



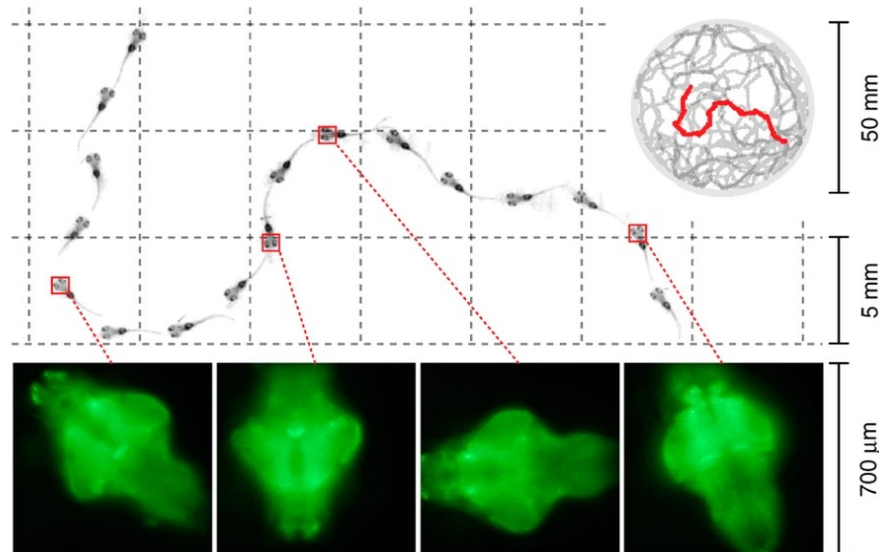
Place Cells in Larval Zebrafish (RoLi Lab, Tübingen)

Tracking Microscopy

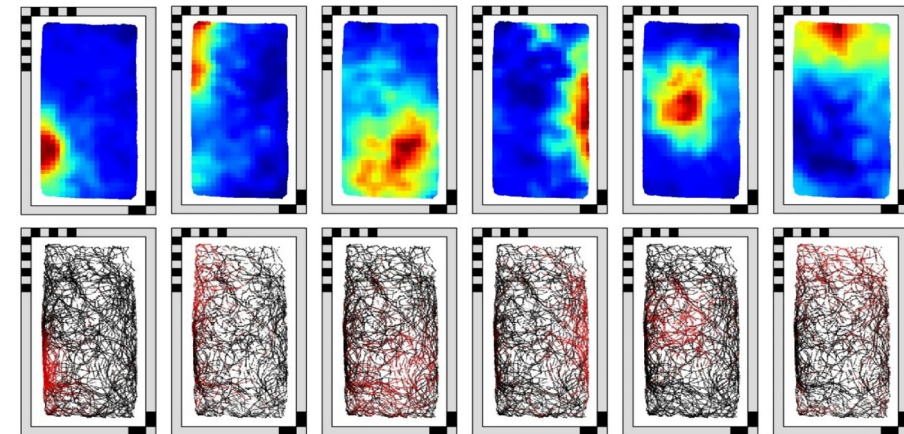


Place Cells in Larval Zebrafish (RoLi Lab, Tübingen)

Tracking Microscopy



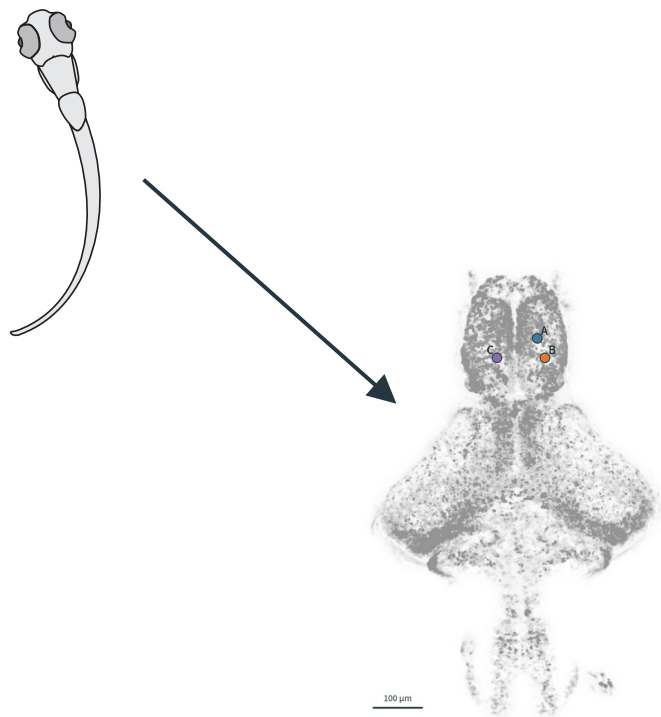
Place Cells



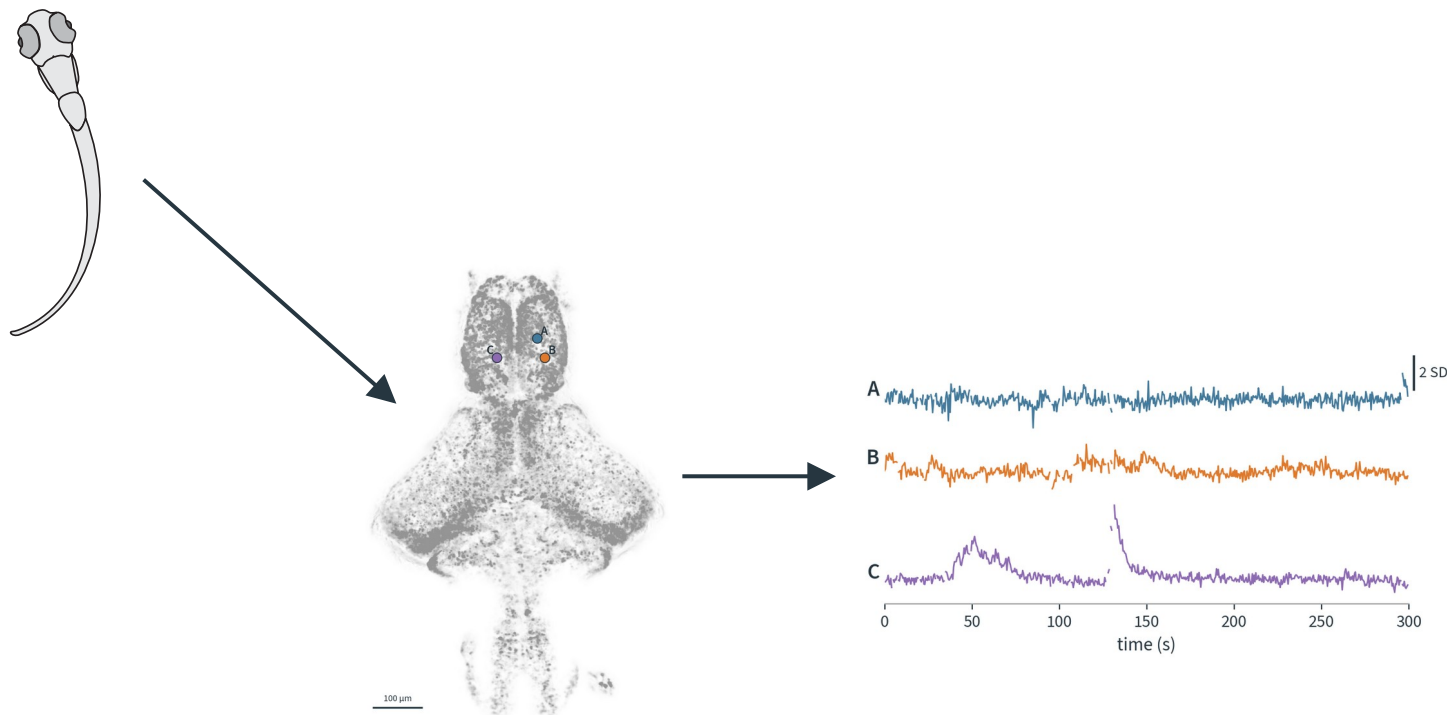
Dynamical Connectomics



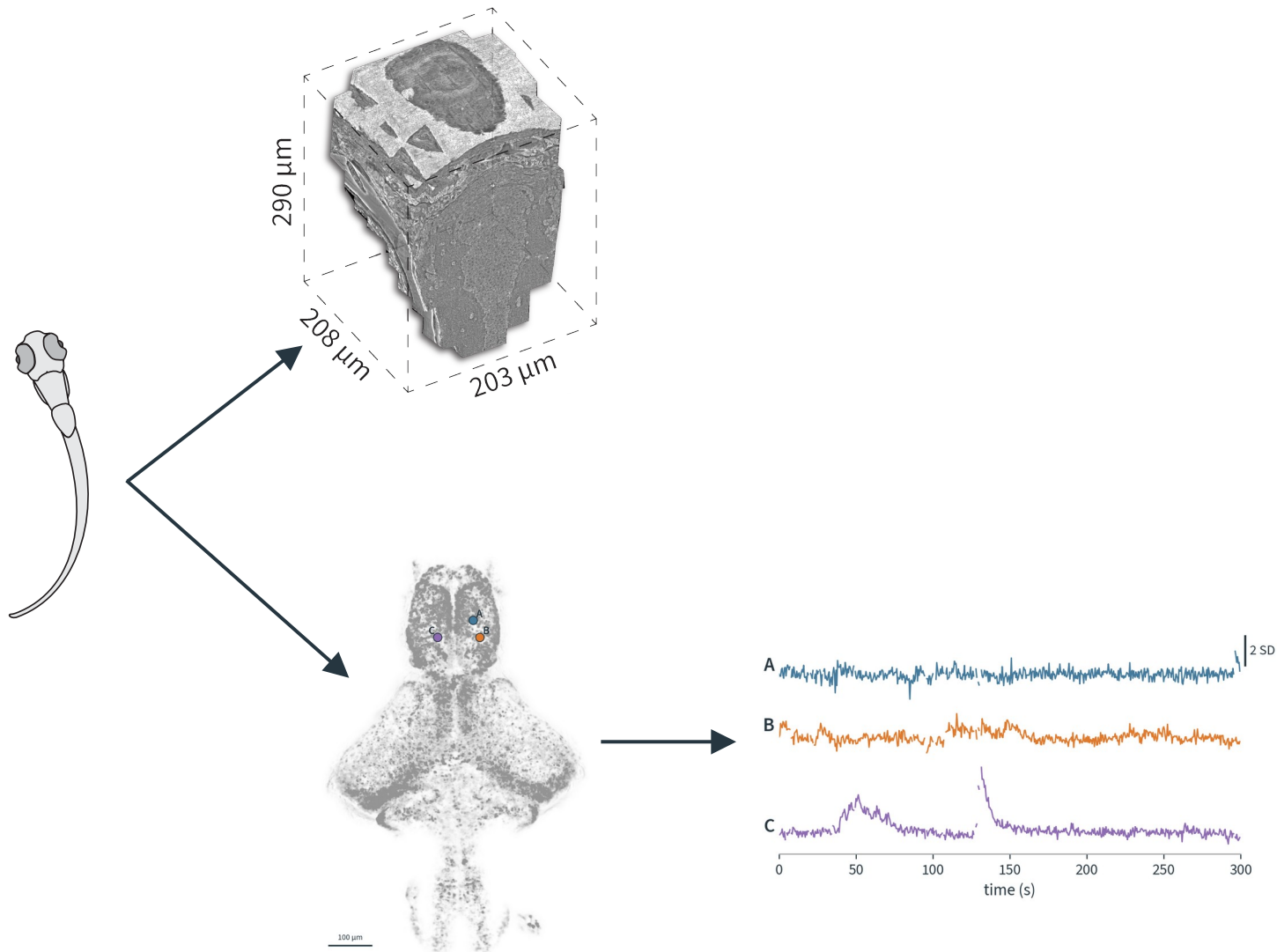
Dynamical Connectomics



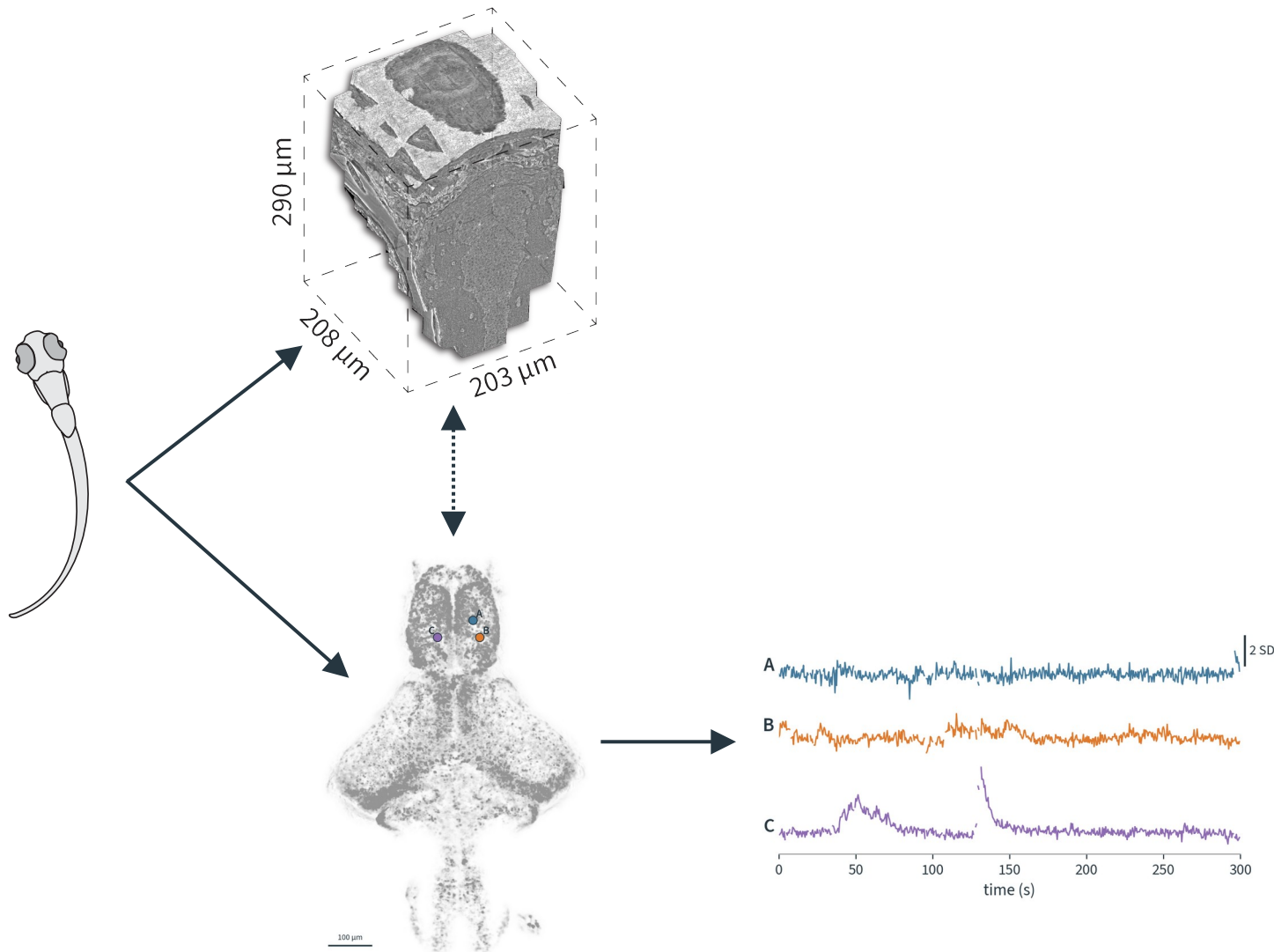
Dynamical Connectomics



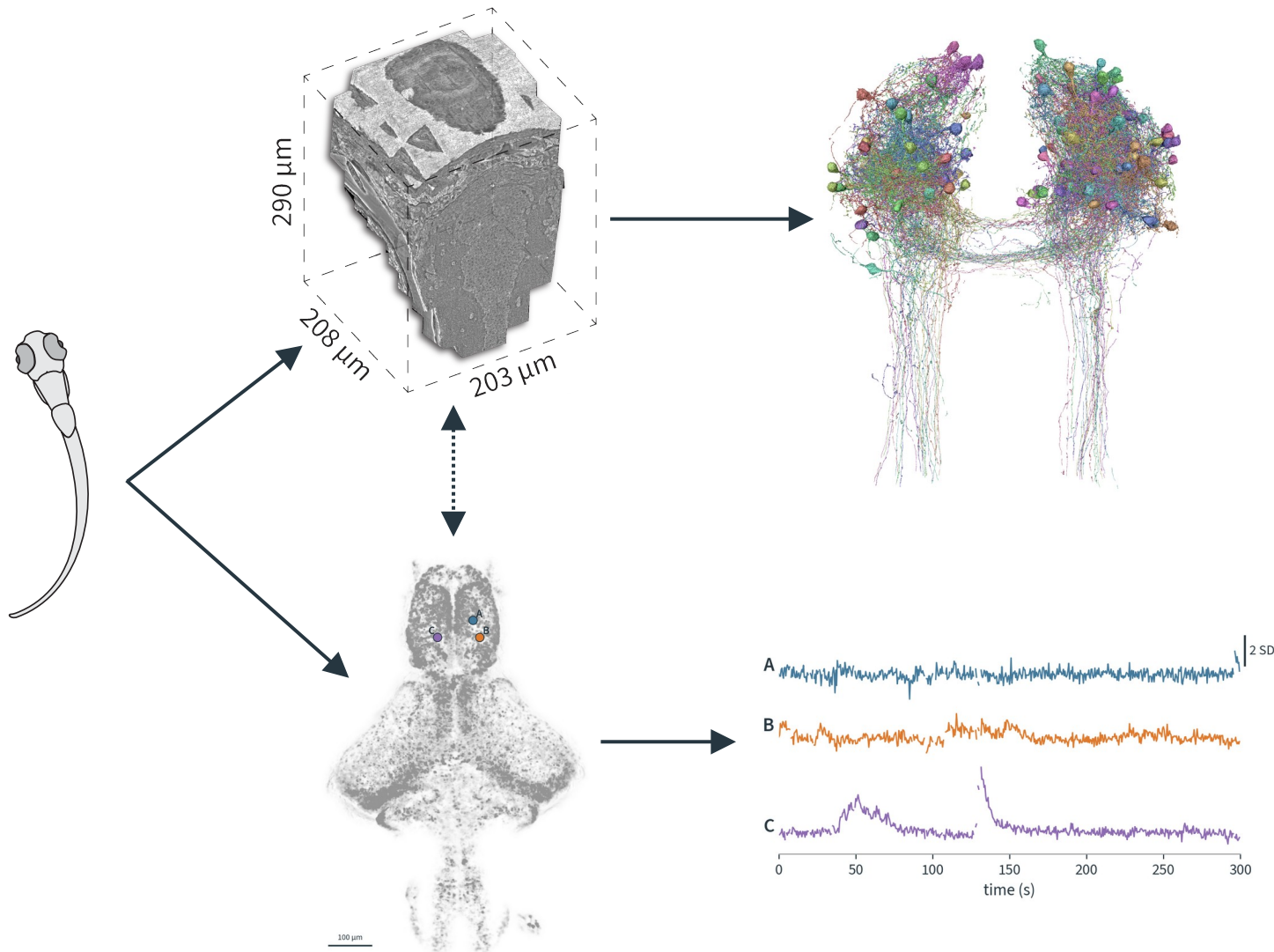
Dynamical Connectomics



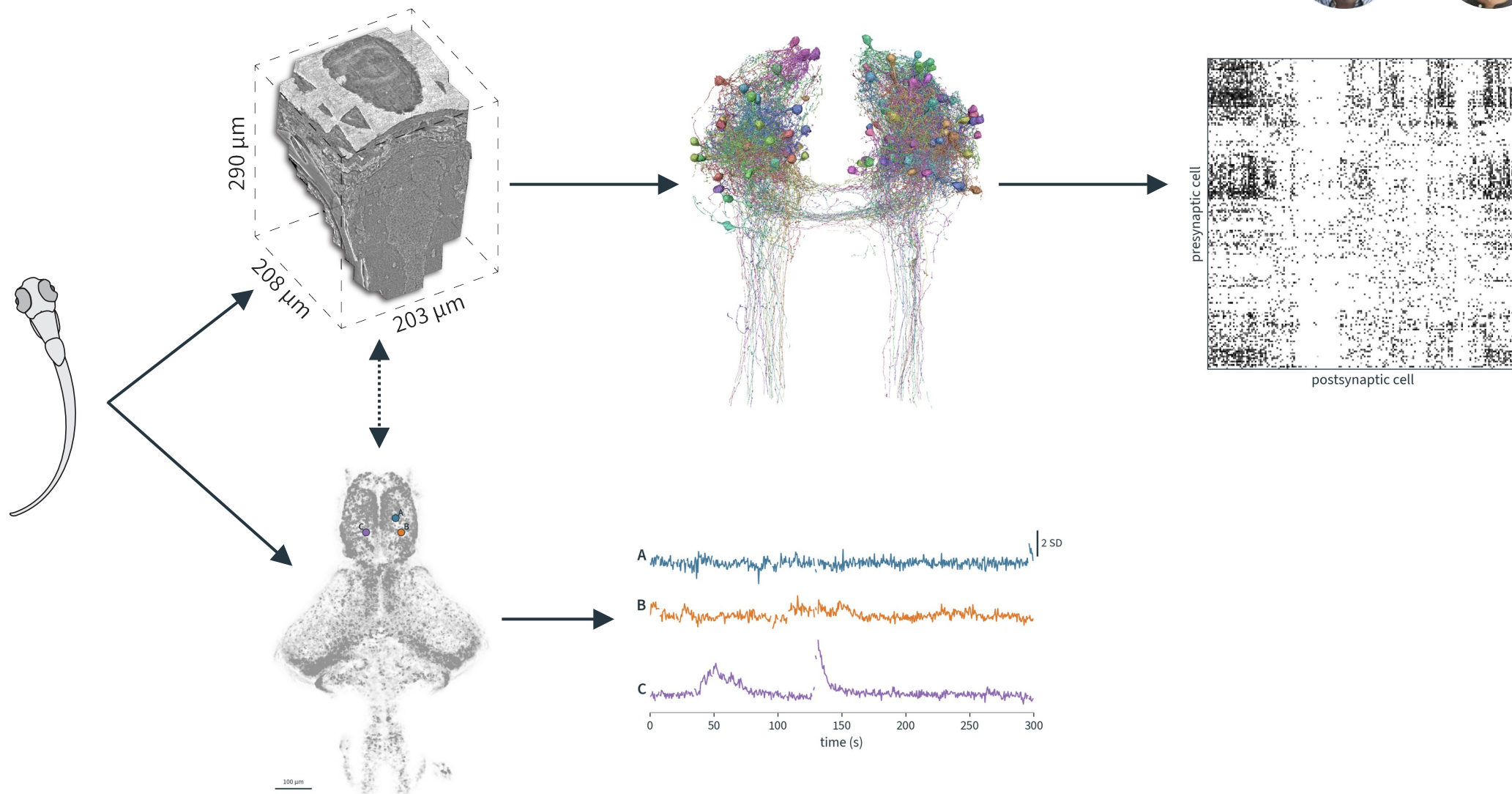
Dynamical Connectomics



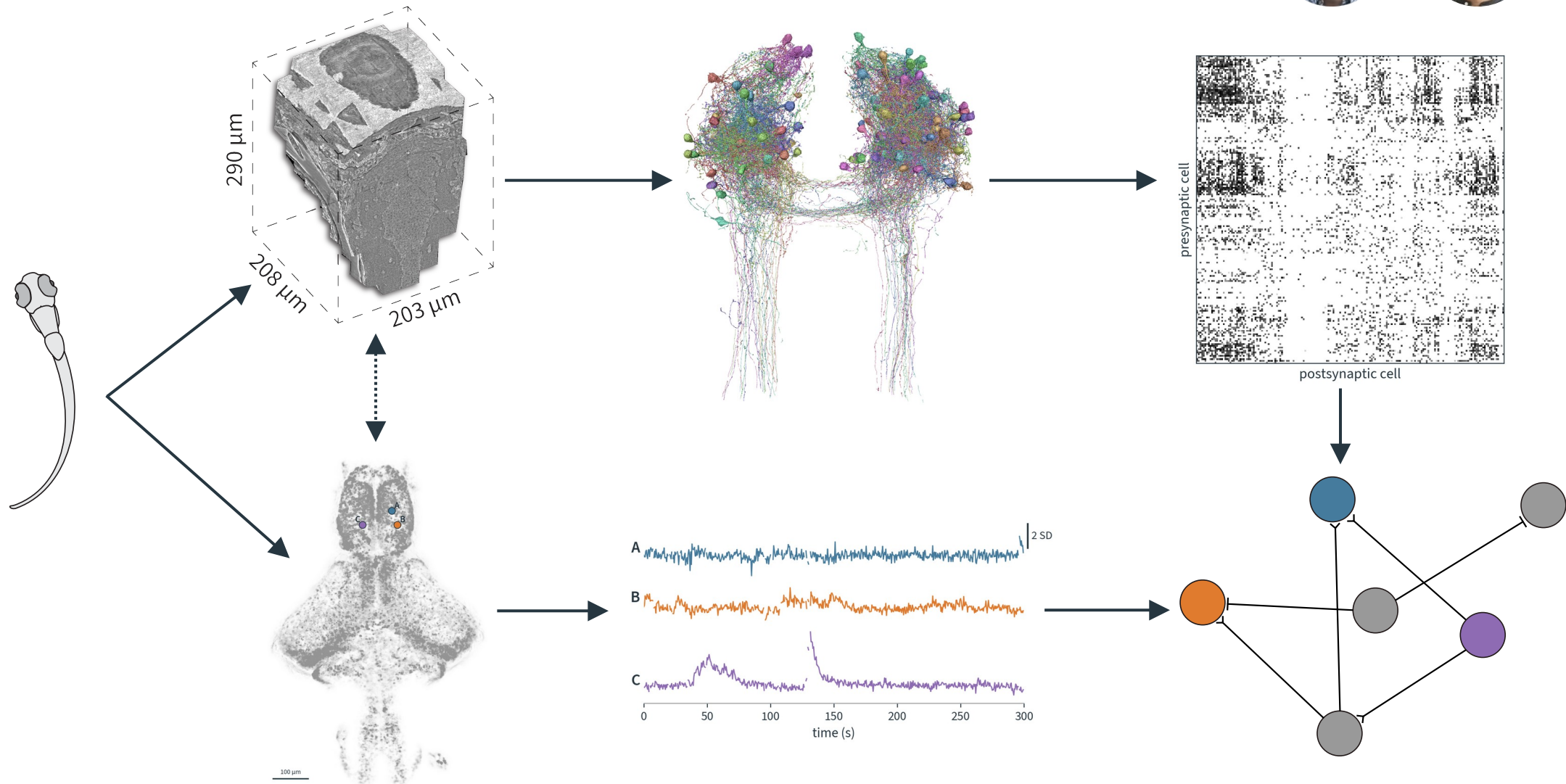
Dynamical Connectomics



Dynamical Connectomics



Dynamical Connectomics



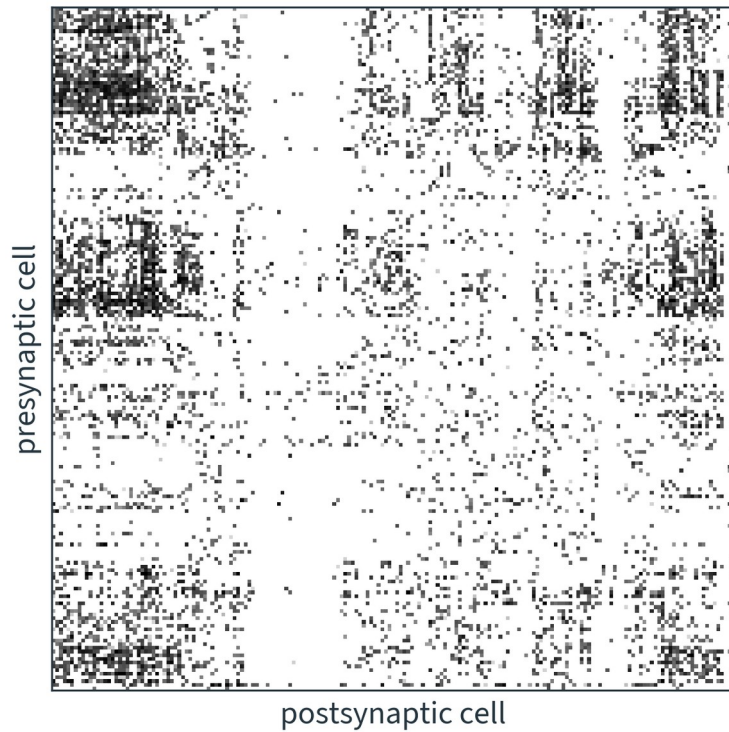
Connectomes Provide Numerous Constraints



Connectomes Provide Numerous Constraints



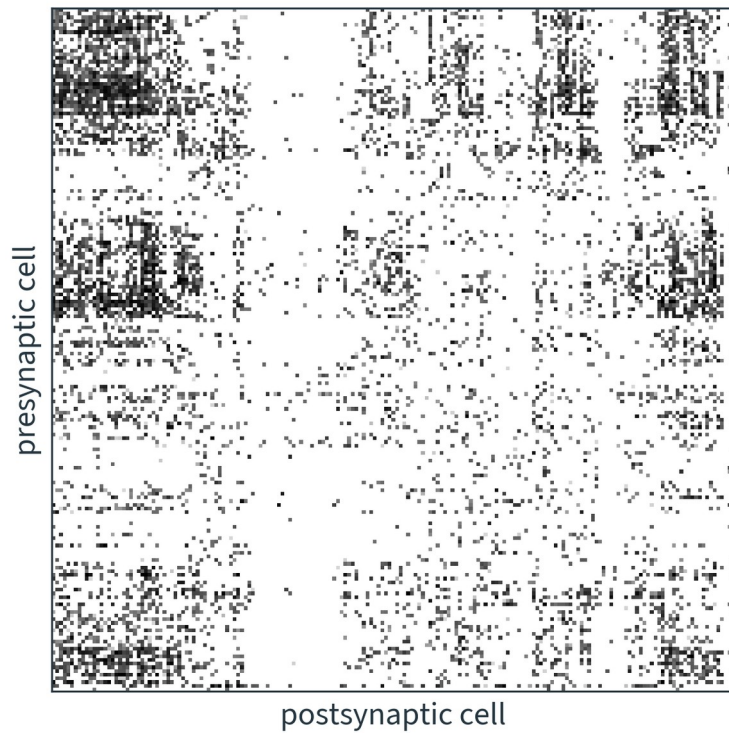
Connectivity



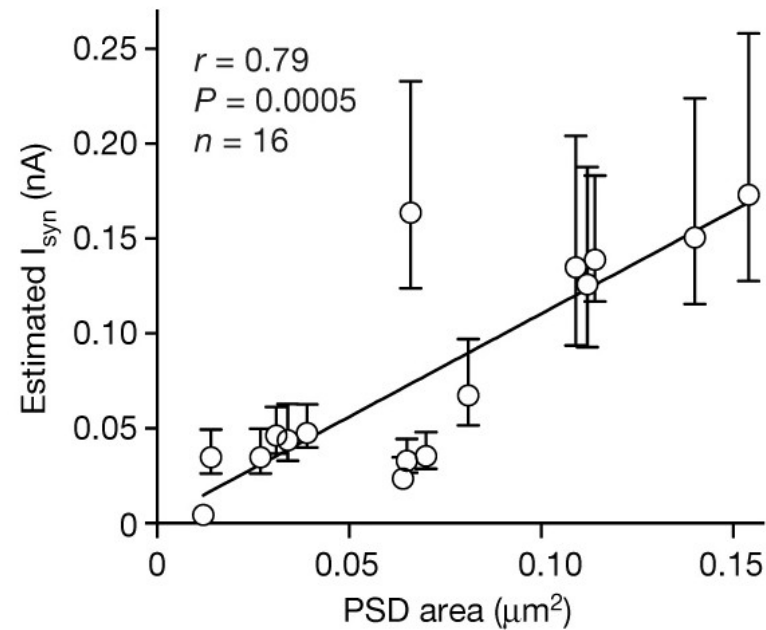
Connectomes Provide Numerous Constraints



Connectivity



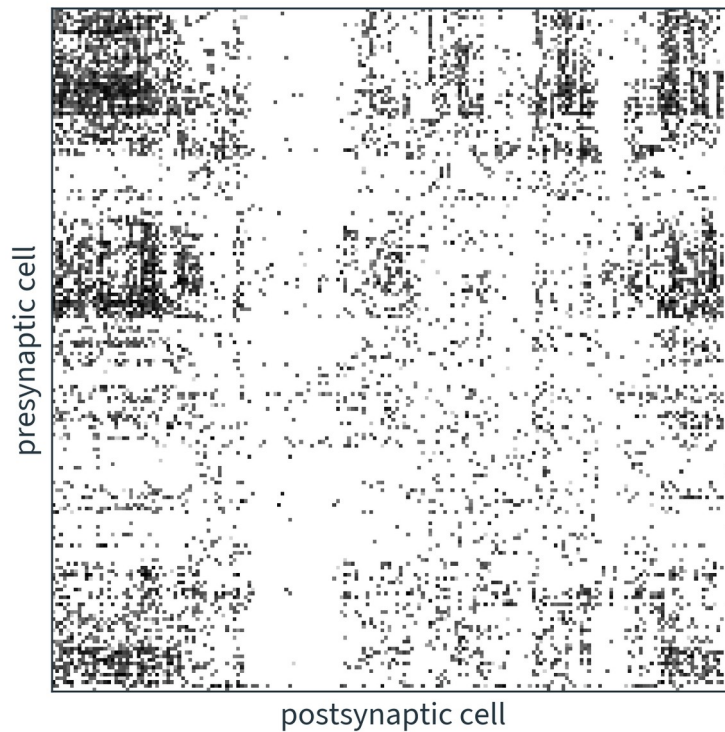
Weight Proxy



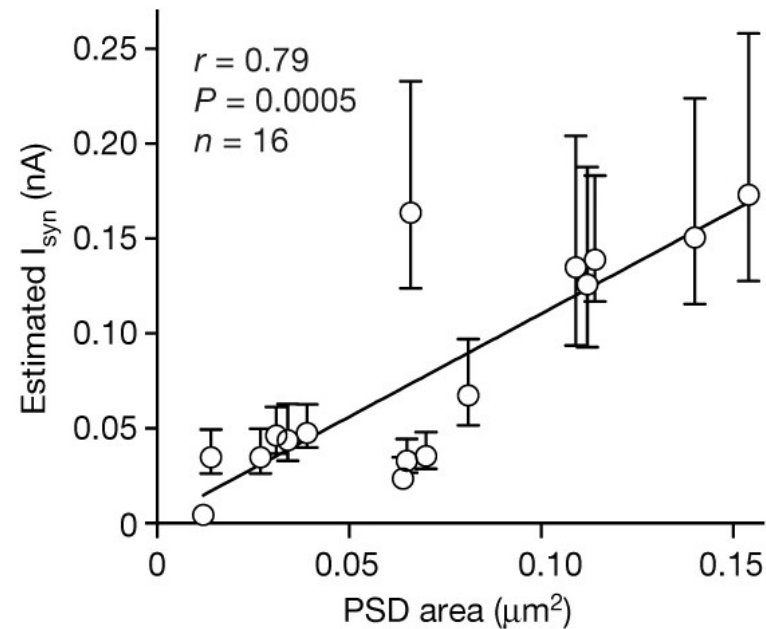
Connectomes Provide Numerous Constraints



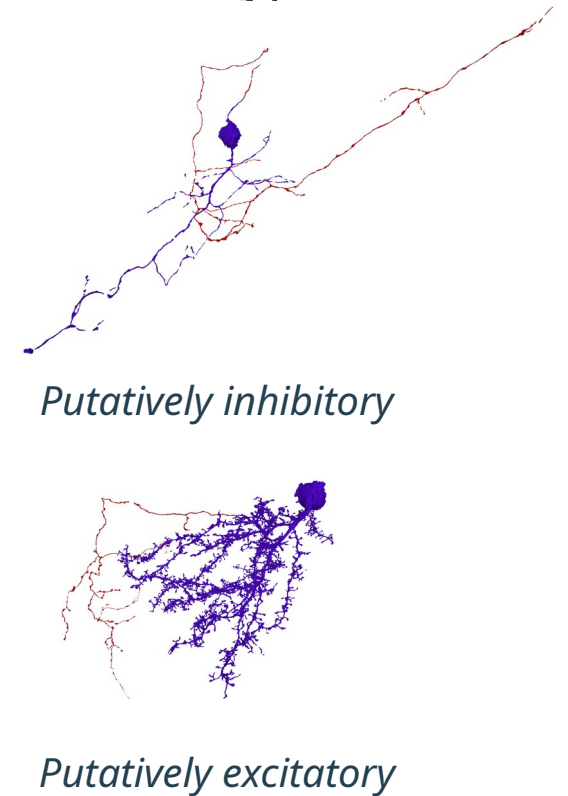
Connectivity



Weight Proxy



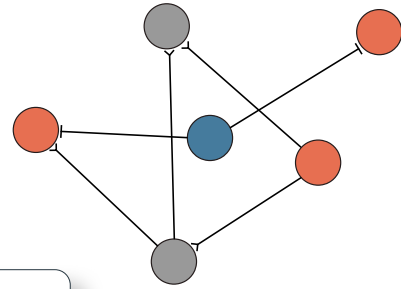
Morphological Cell Types



Real Data Comes With Imperfections

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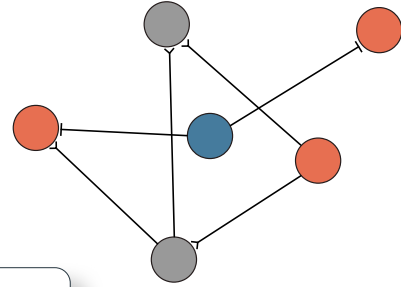
1) Unrecorded activity



~15% of EM somata have registered calcium traces

Real Data Comes With Imperfections

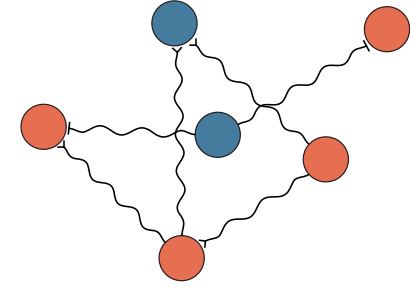
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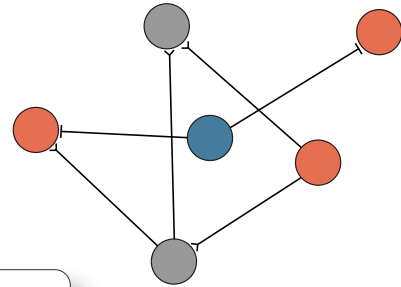
2) Noisy weight estimates

$r \approx 0.79$ correlation
between synaptic volume
and weight



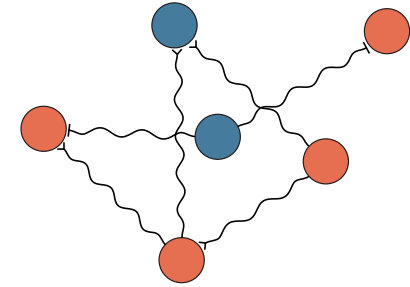
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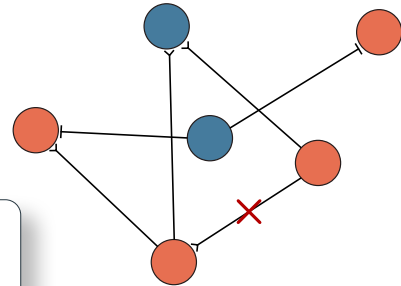
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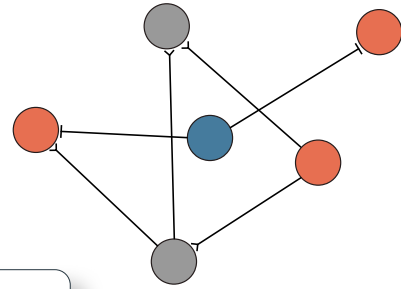
3) Reconstruction errors



Estimate $\geq 15\%$ synapses missing in proofread neurons

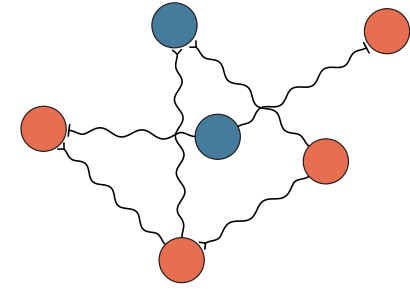
Real Data Comes With Imperfections

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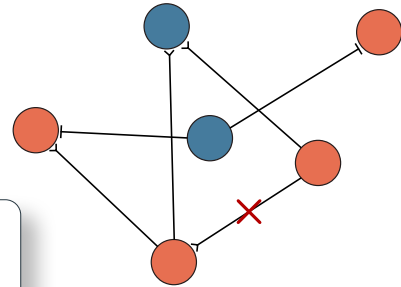
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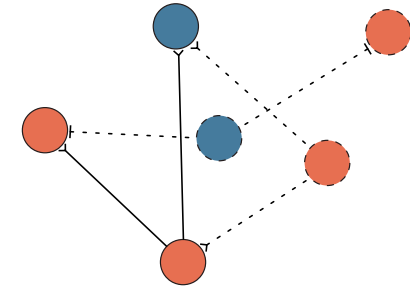
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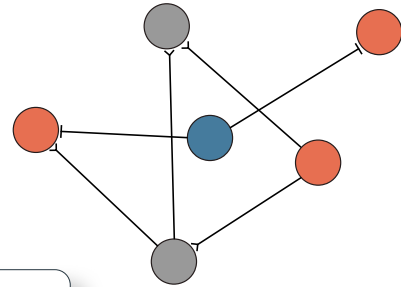
4) Partial reconstruction



~7% of the ~10,000 telencephalic neurons reconstructed (~1,100 place cells)

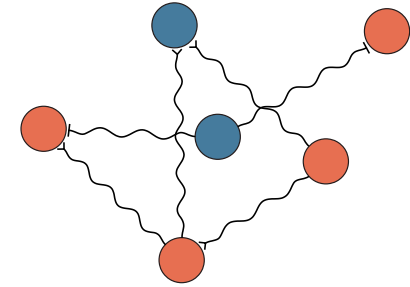
Real Data Comes With Imperfections

1) Unrecorded activity



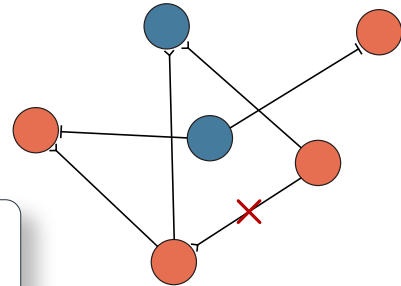
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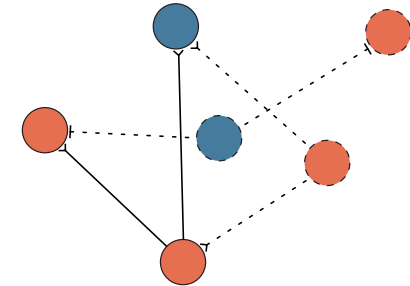
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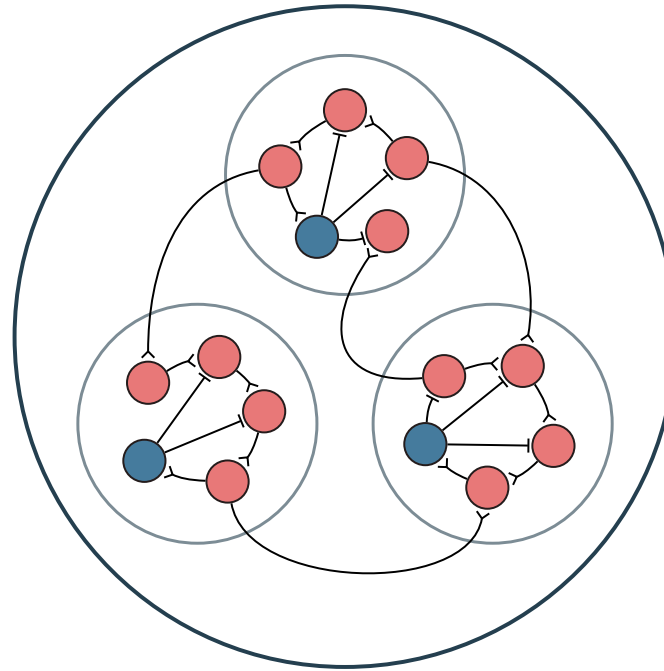
Question: Can a mechanistic network model survive realistic data imperfections?

Part 1: Elucidating the implementation of a spatial cognitive map using connectome-constrained neural network models

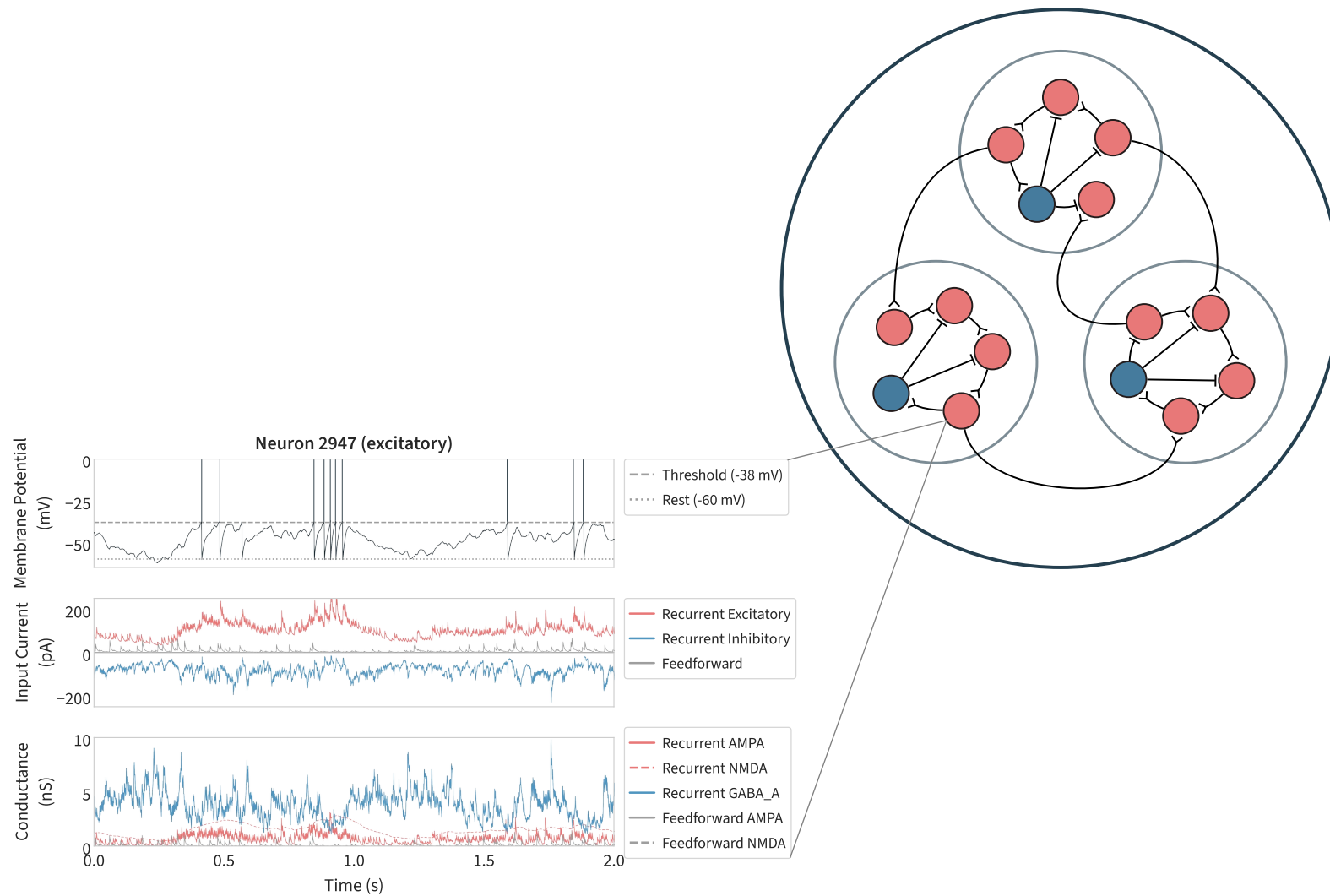
Part 2: Robustness of circuit fits to realistic connectome degradations in a teacher-student learning paradigm

Hand-Crafting a Spiking Teacher Network

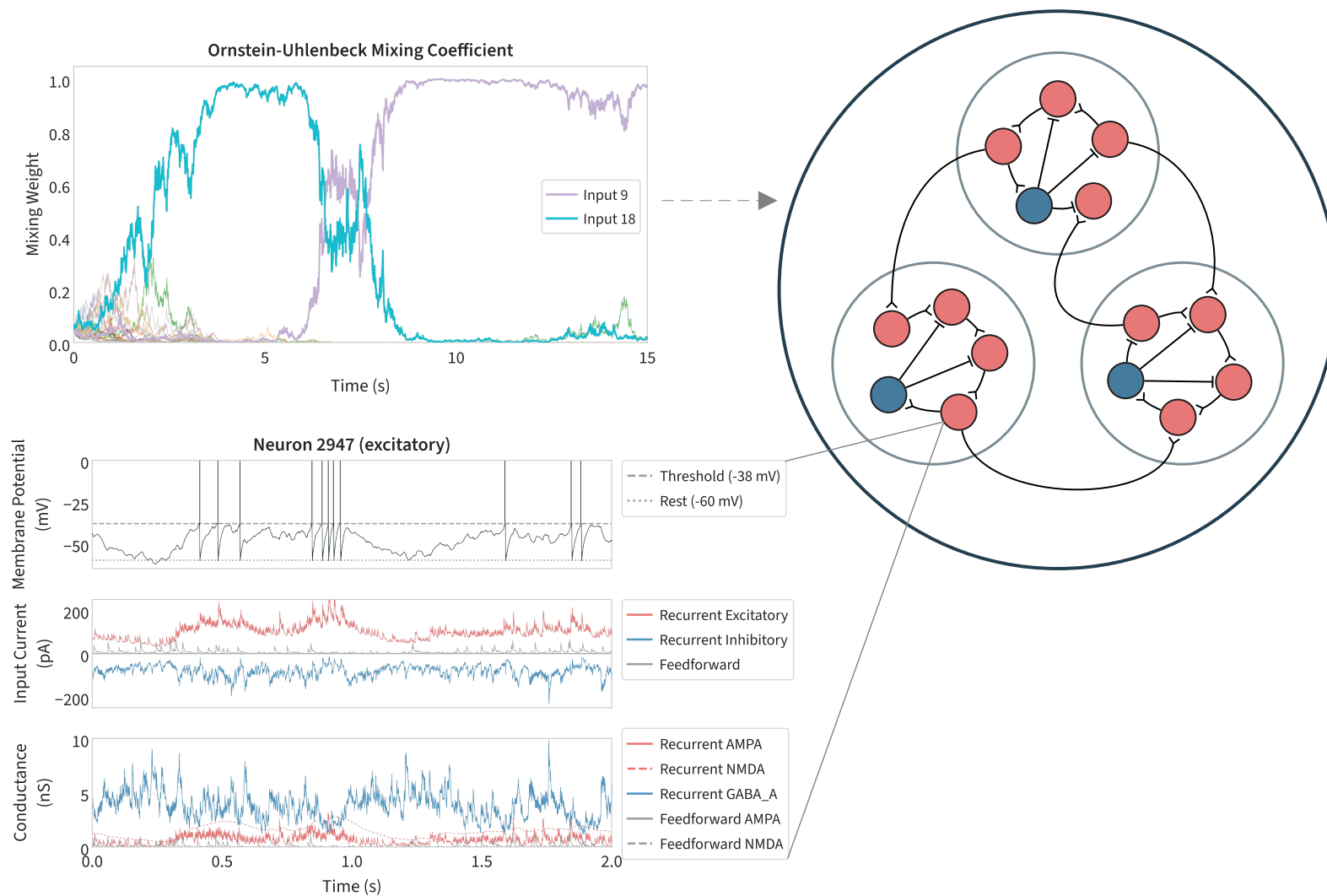
Hand-Crafting a Spiking Teacher Network



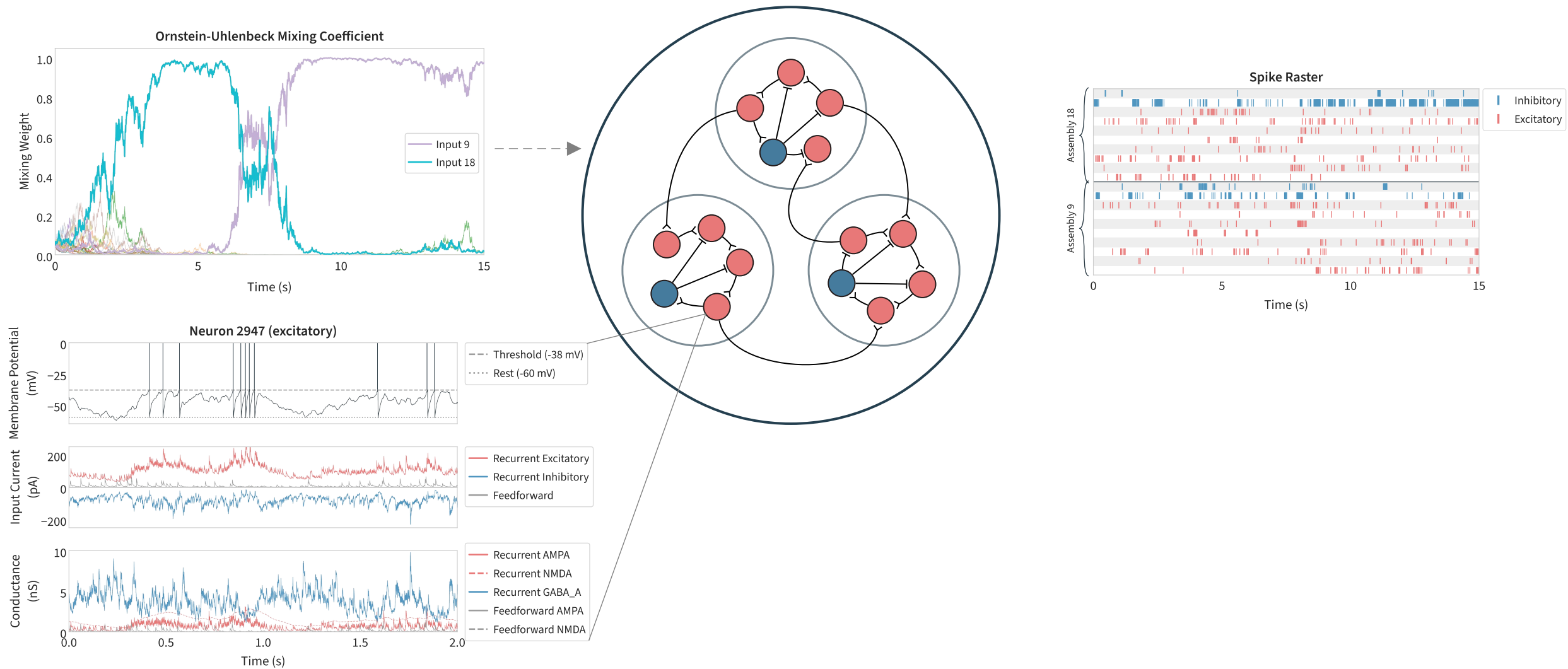
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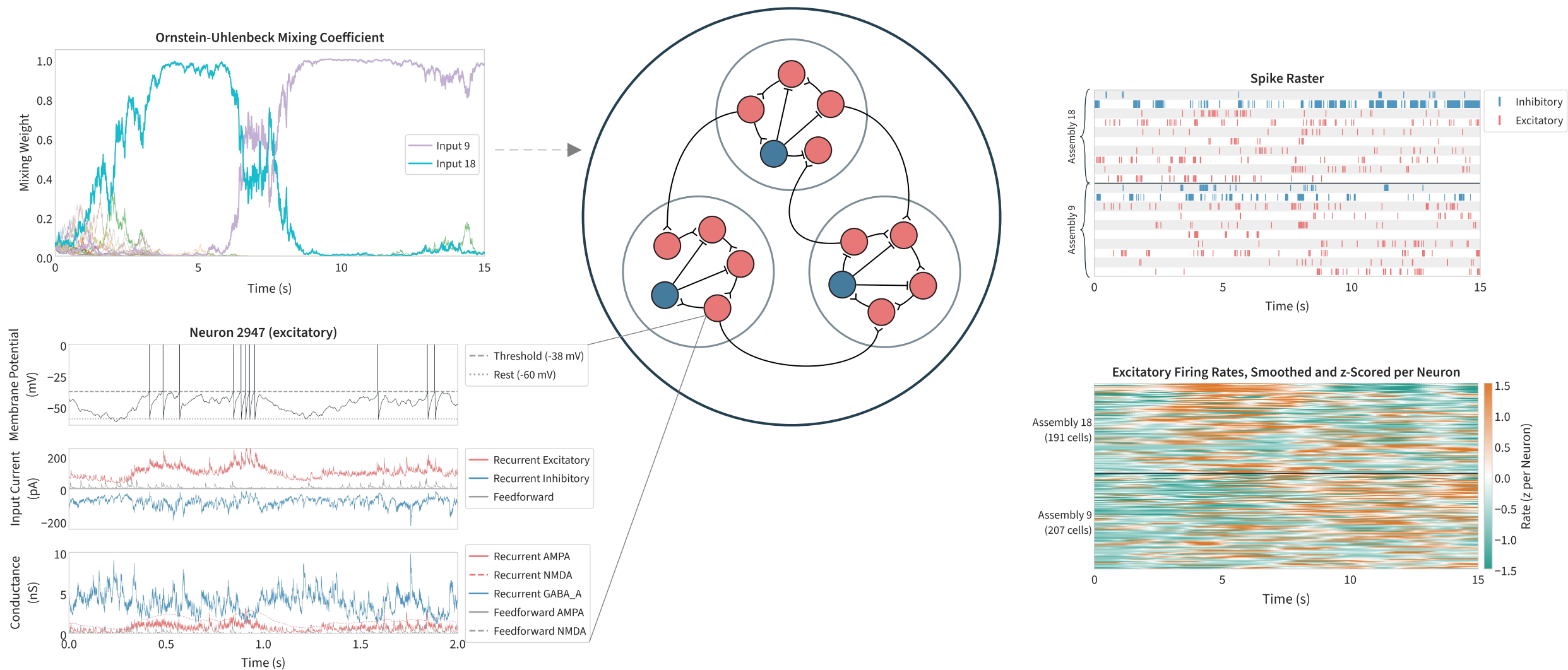
Hand-Crafting a Spiking Teacher Network



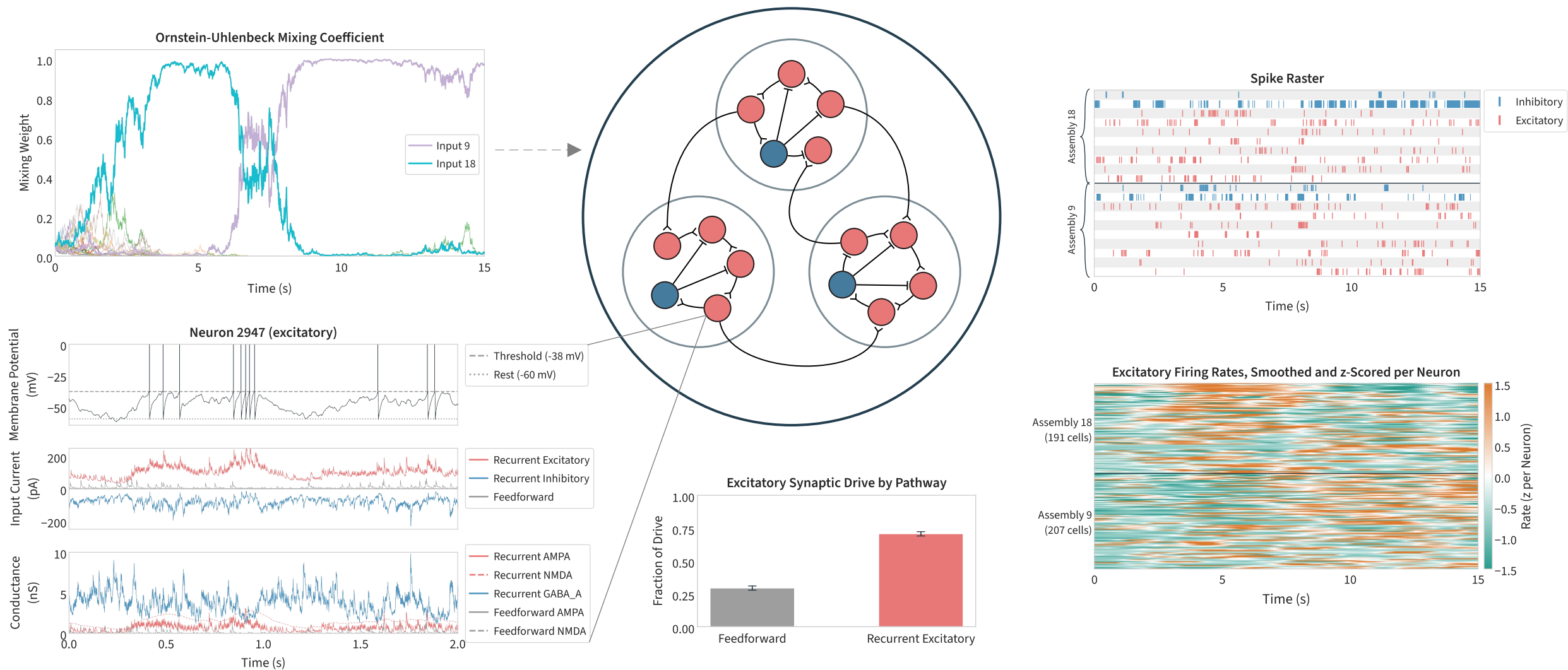
Hand-Crafting a Spiking Teacher Network



Hand-Crafting a Spiking Teacher Network

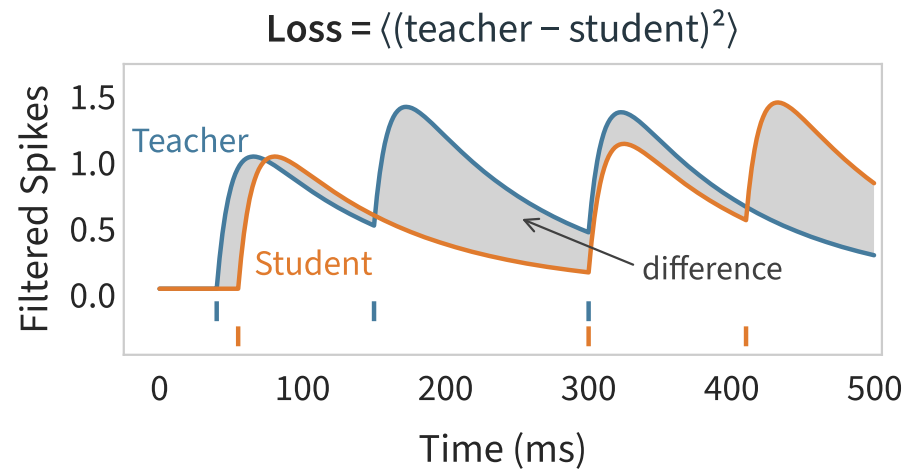
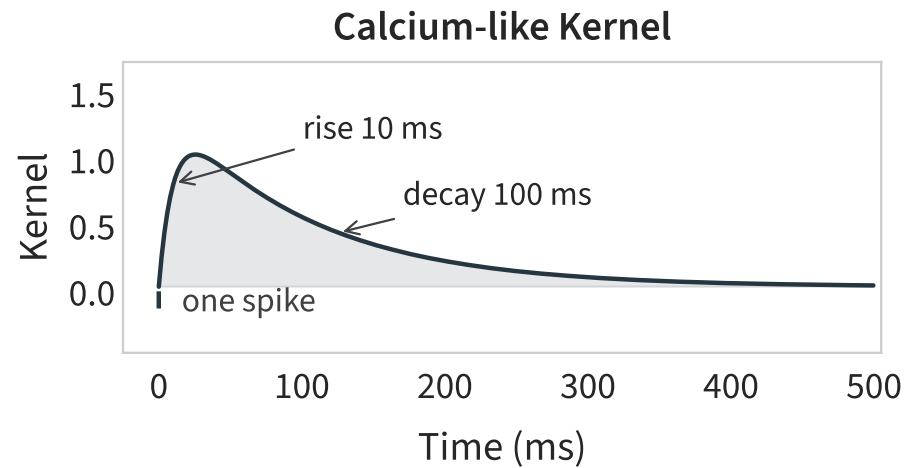


Hand-Crafting a Spiking Teacher Network

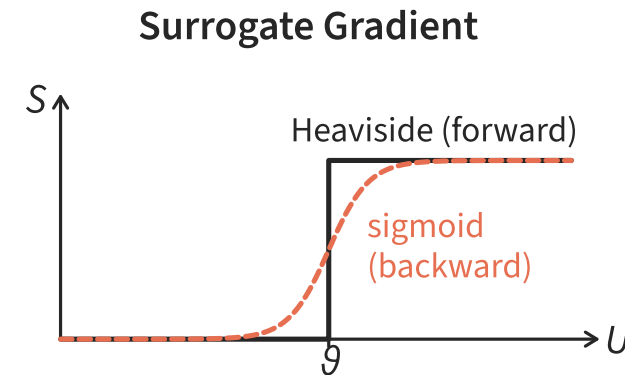
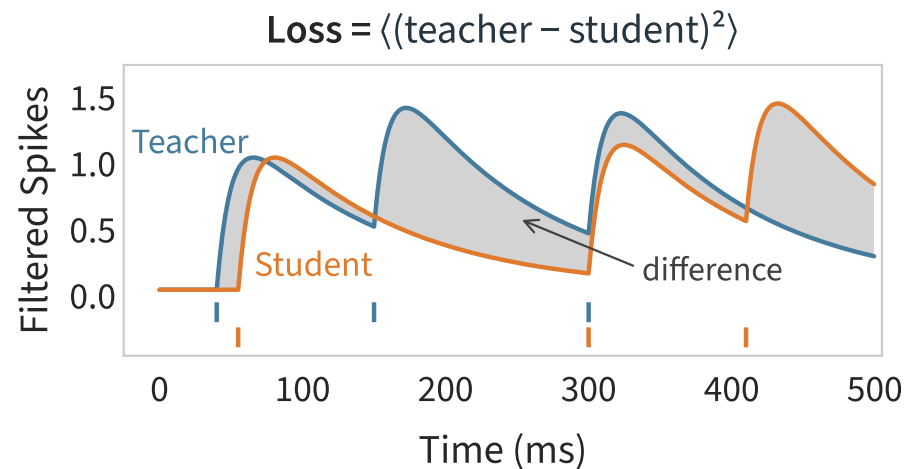
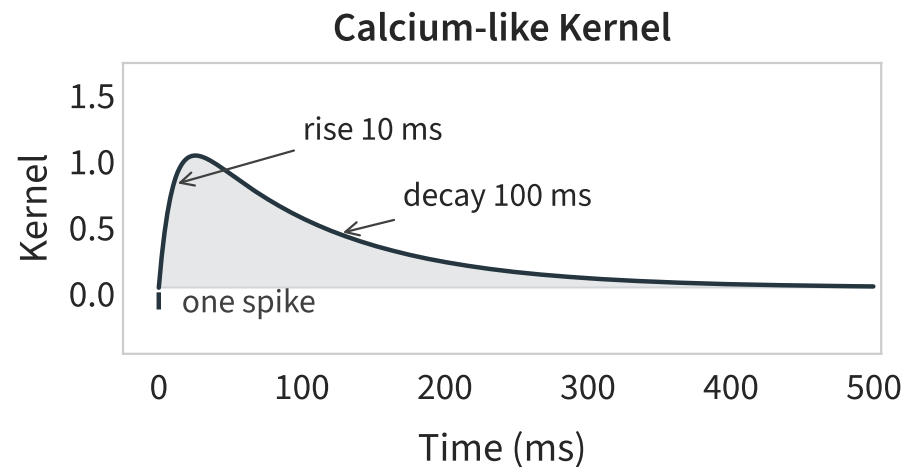


Training Student to Match the Teacher

Training Student to Match the Teacher

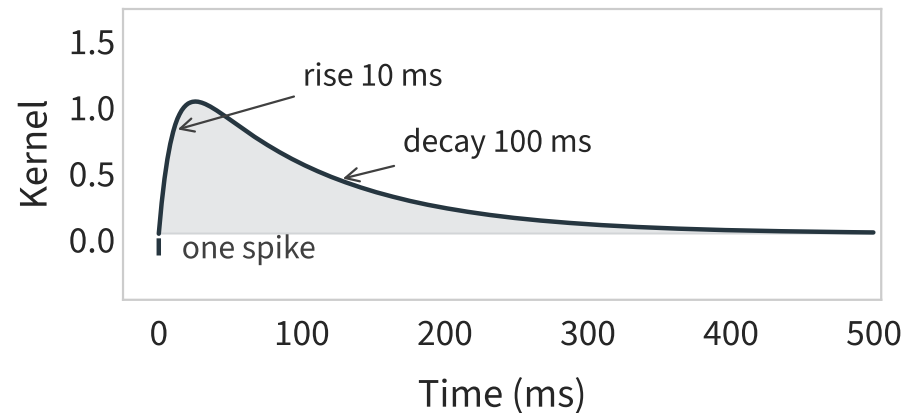


Training Student to Match the Teacher

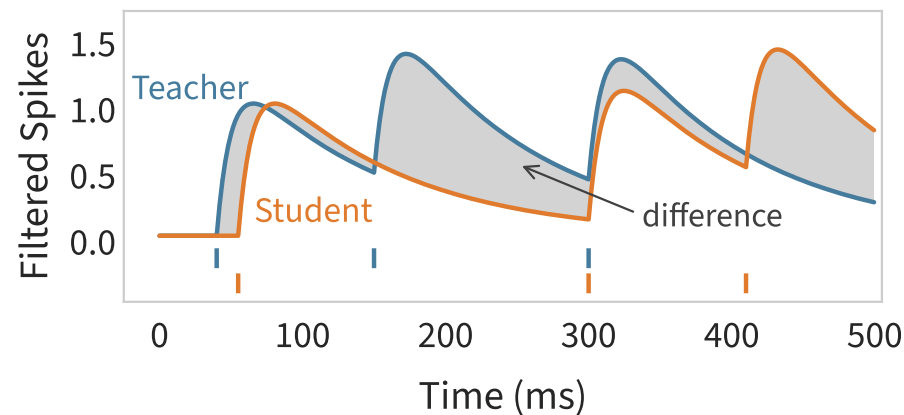


Training Student to Match the Teacher

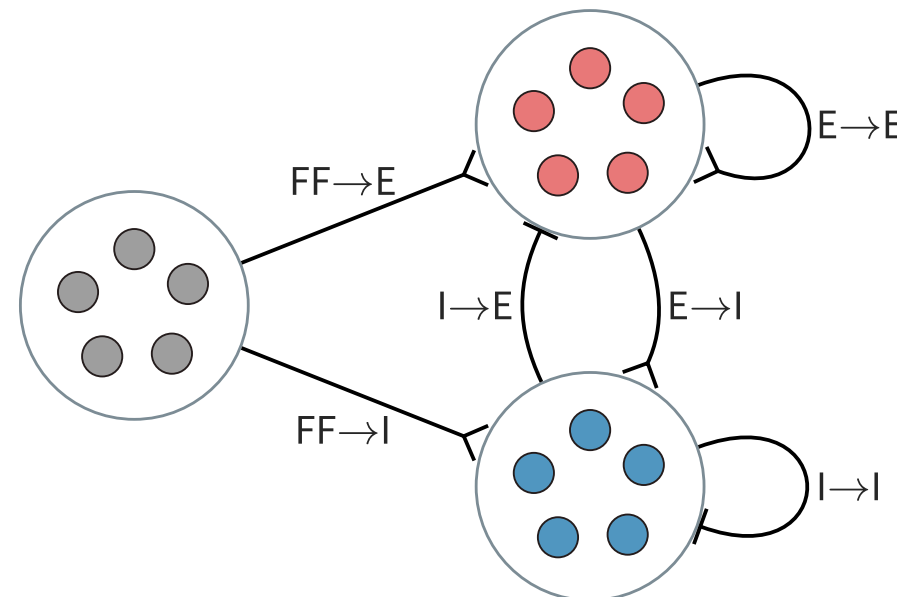
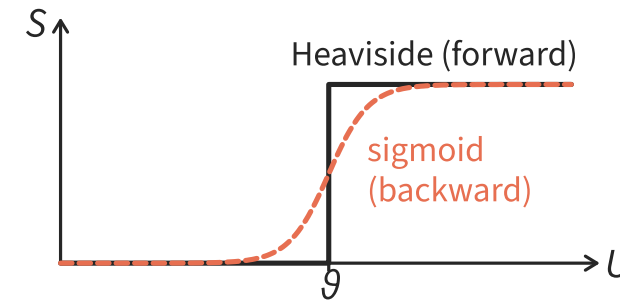
Calcium-like Kernel



$$\text{Loss} = \langle (\text{teacher} - \text{student})^2 \rangle$$



Surrogate Gradient

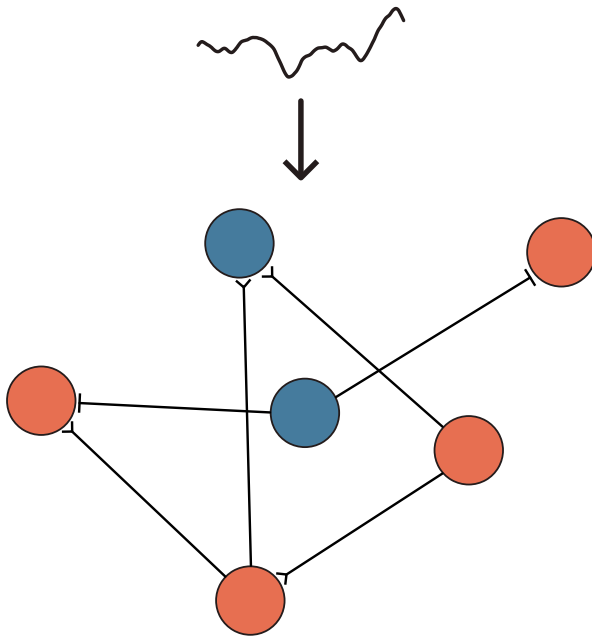


- Assume correct neuron model
- Optimise 6 scaling factors

Evaluating a Circuit Fit

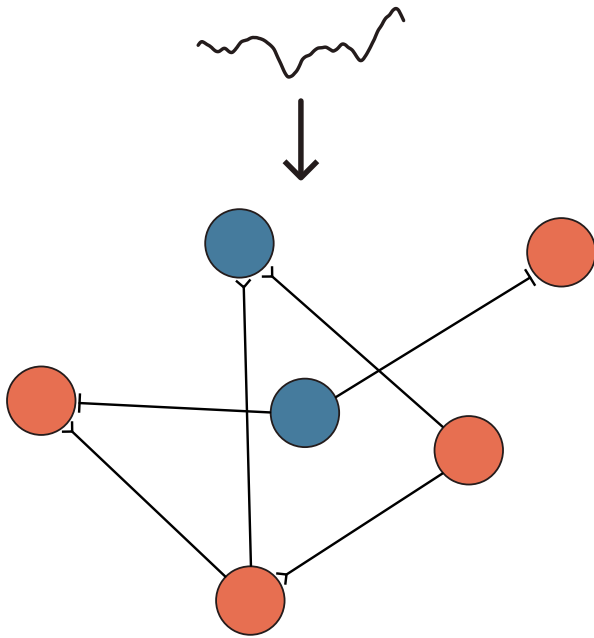
Evaluating a Circuit Fit

Generalise to
Unseen Inputs

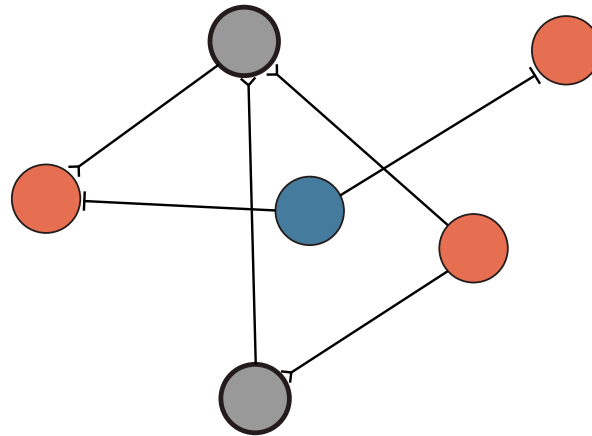


Evaluating a Circuit Fit

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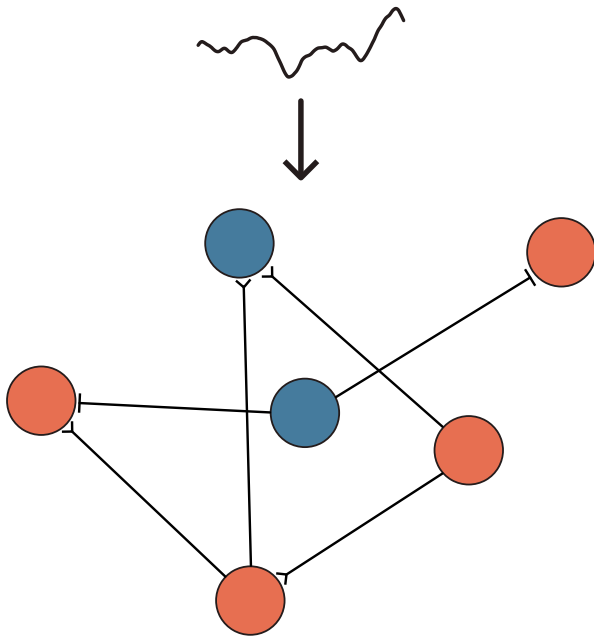


Prediction of
Unrecorded Neurons

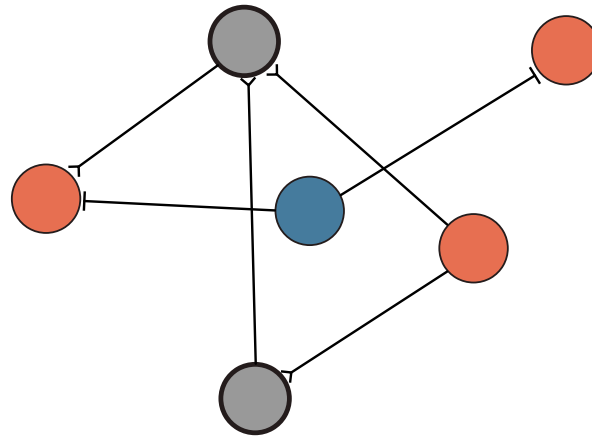


Evaluating a Circuit Fit

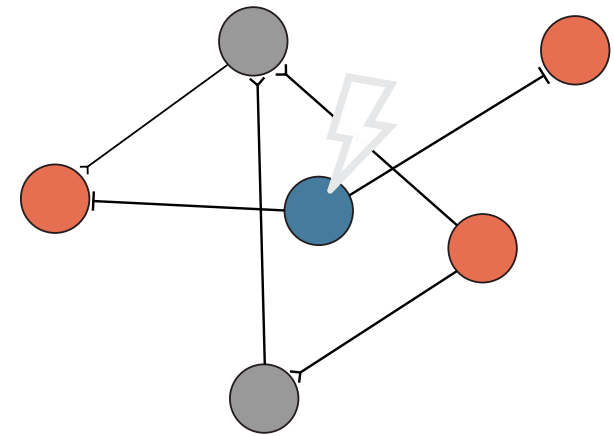
Generalise to
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Prediction of
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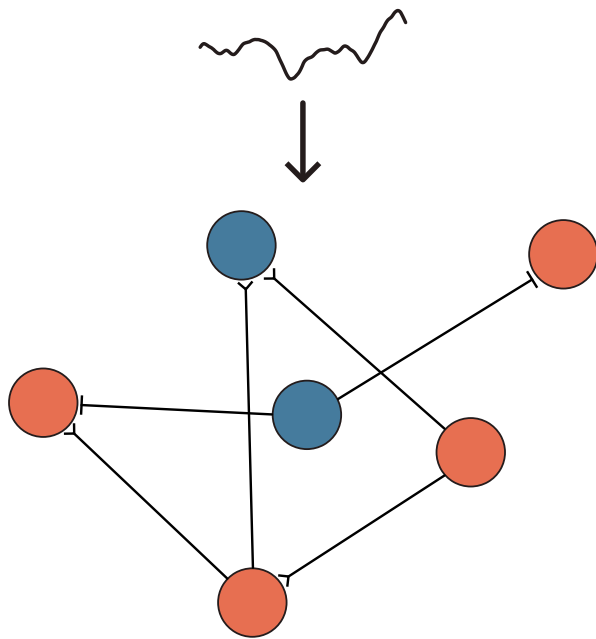


Robustness to
Perturbations

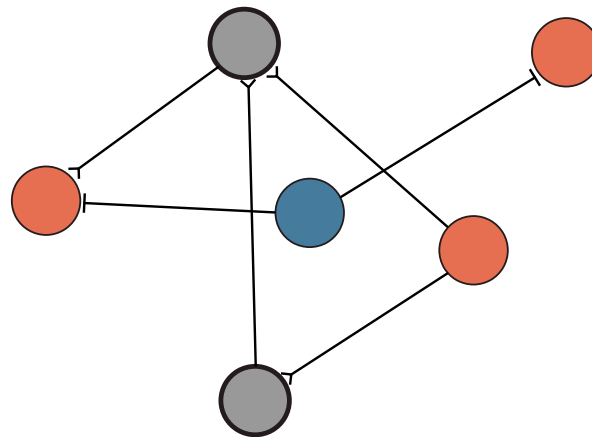


Evaluating a Circuit Fit

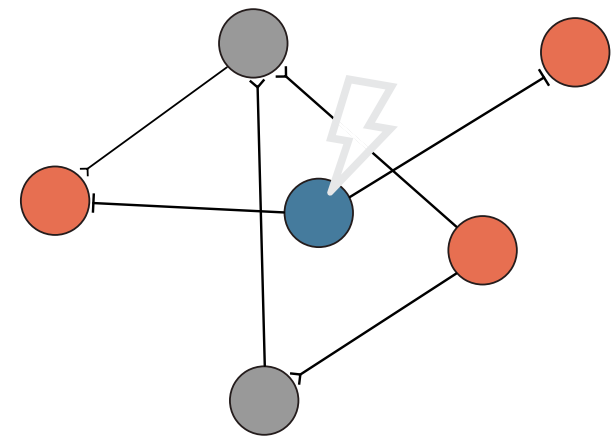
Generalise to
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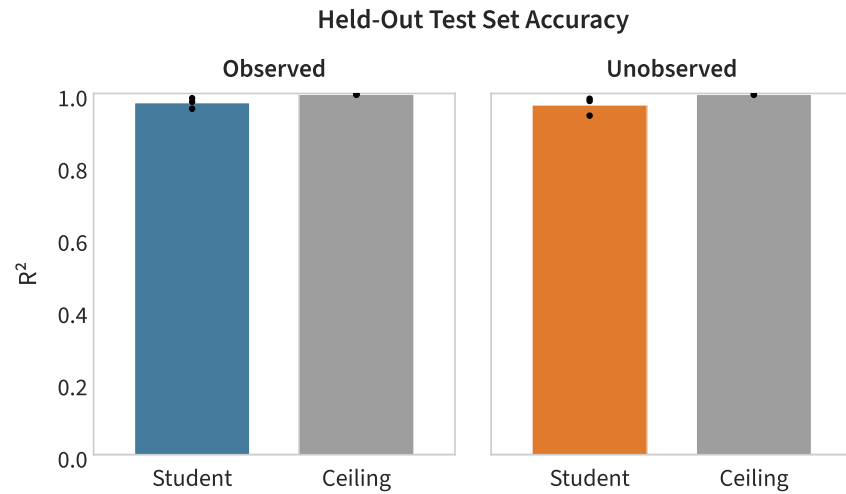
Robustness to
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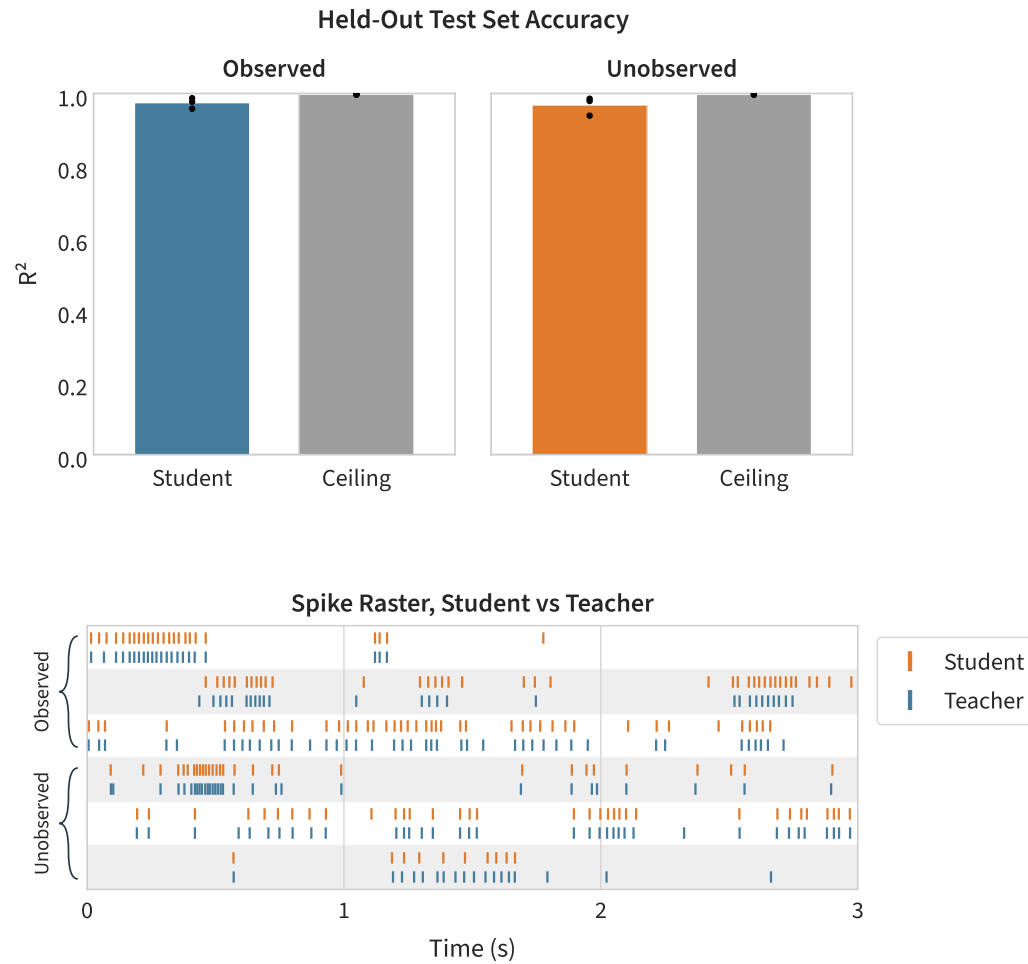
Inject hyperpolarising current into
25% of inhibitory neurons,
reducing their firing rate by 50%

Can a Student Match a Teacher with 50% Neurons Observed?

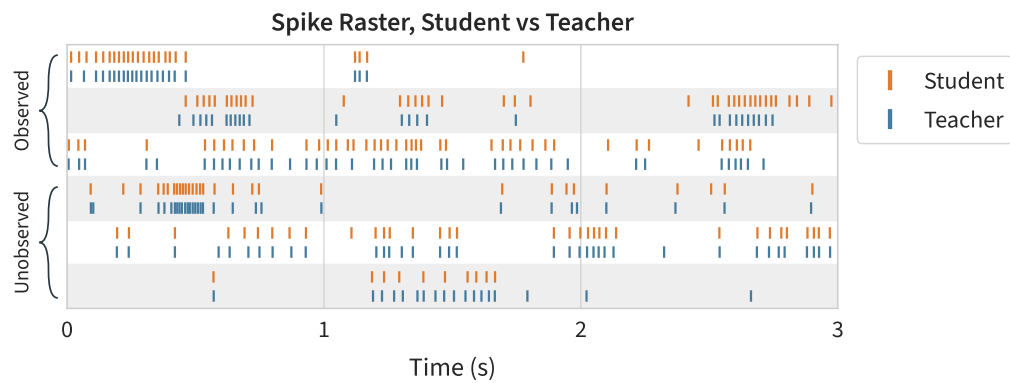
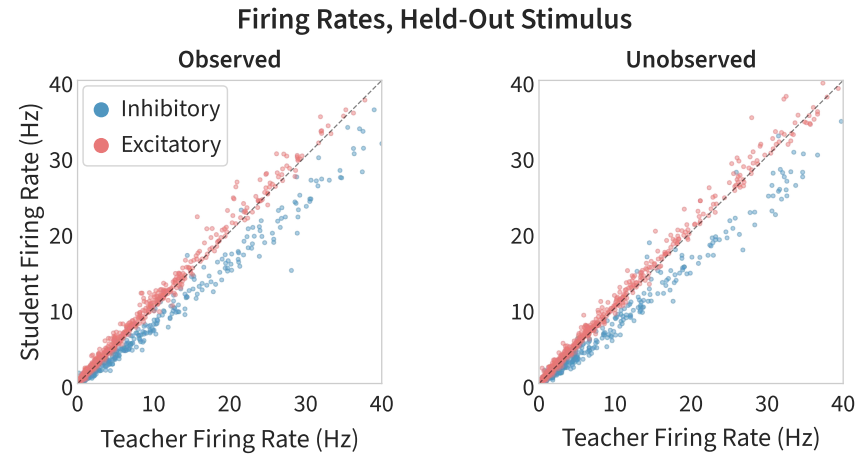
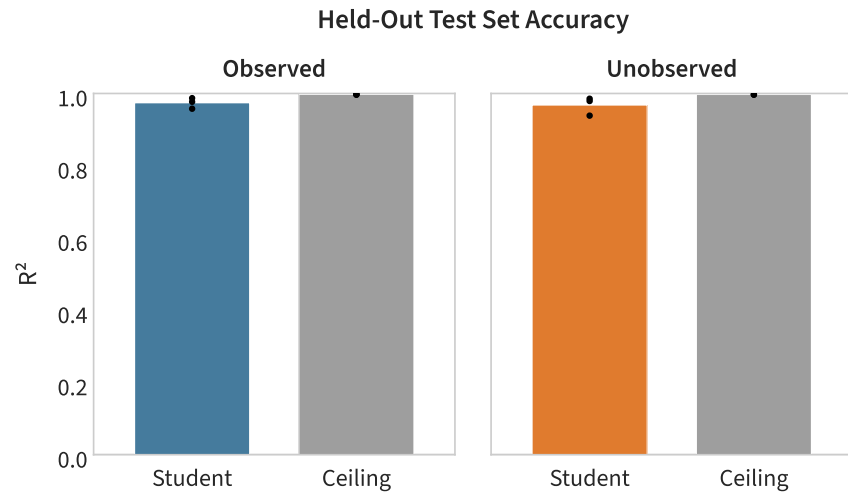
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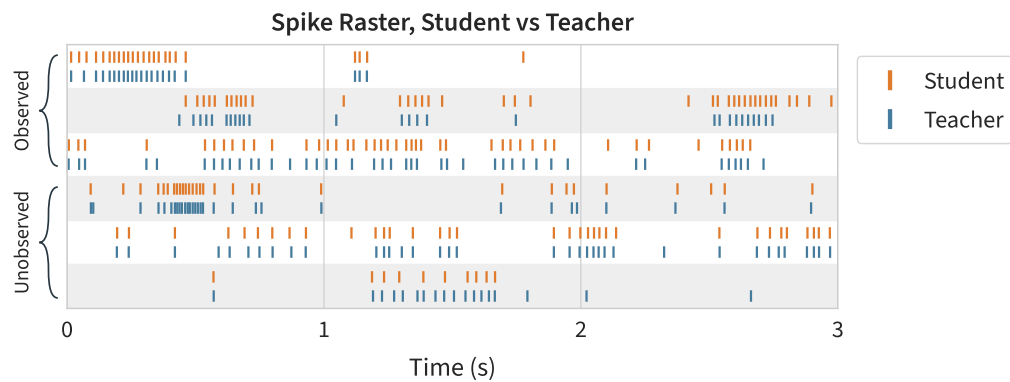
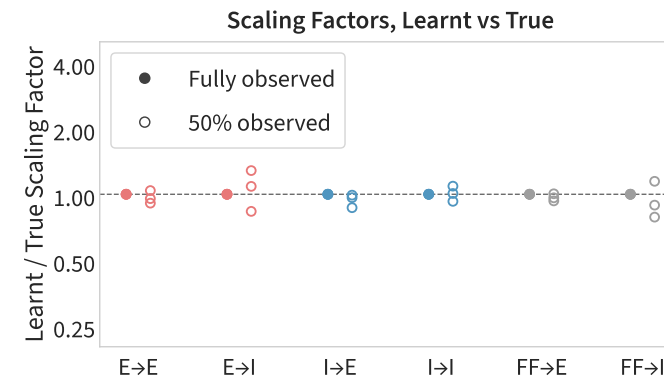
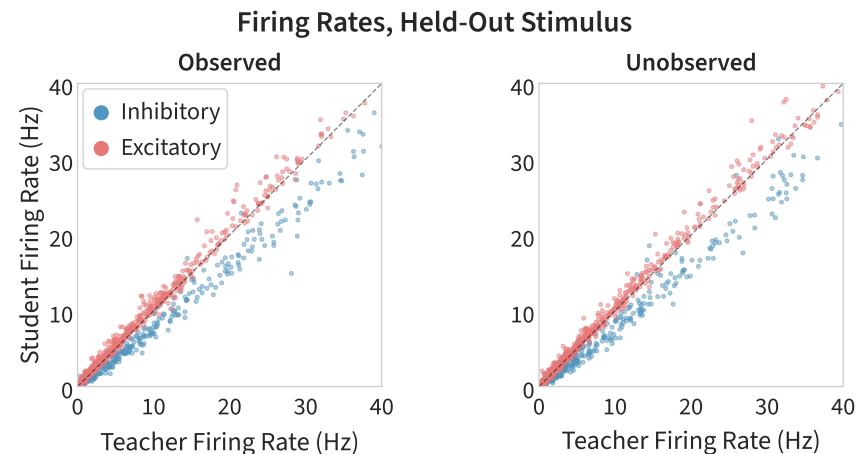
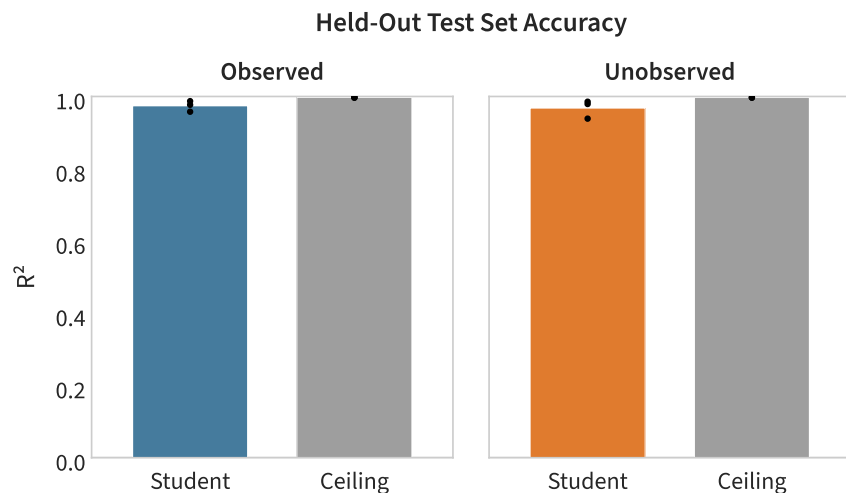
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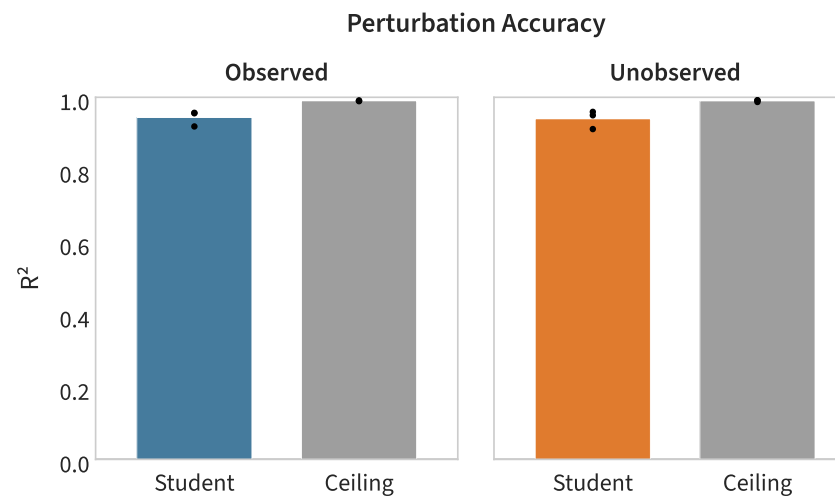
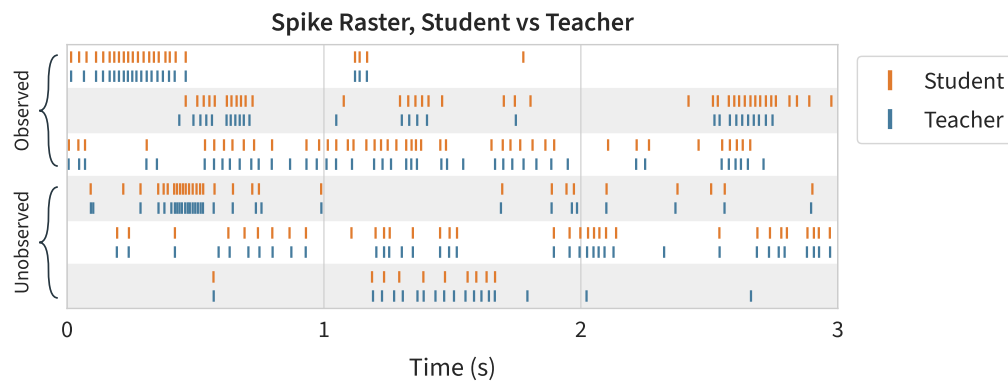
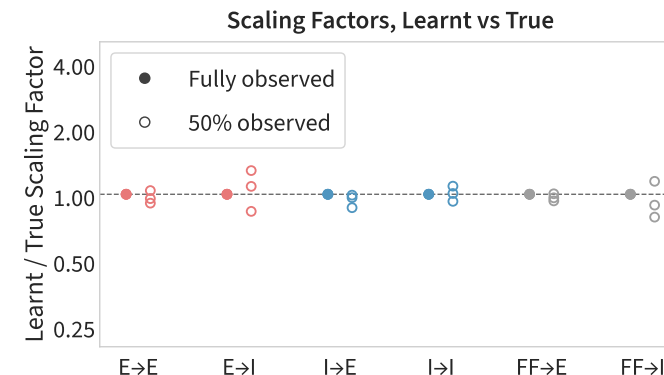
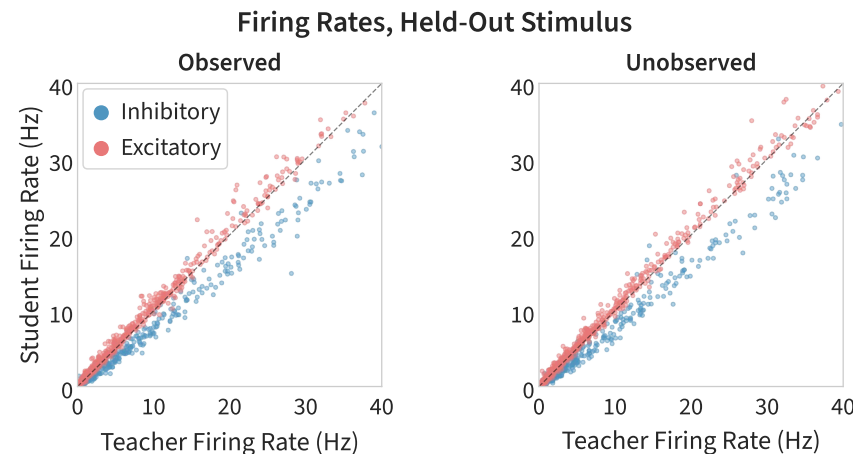
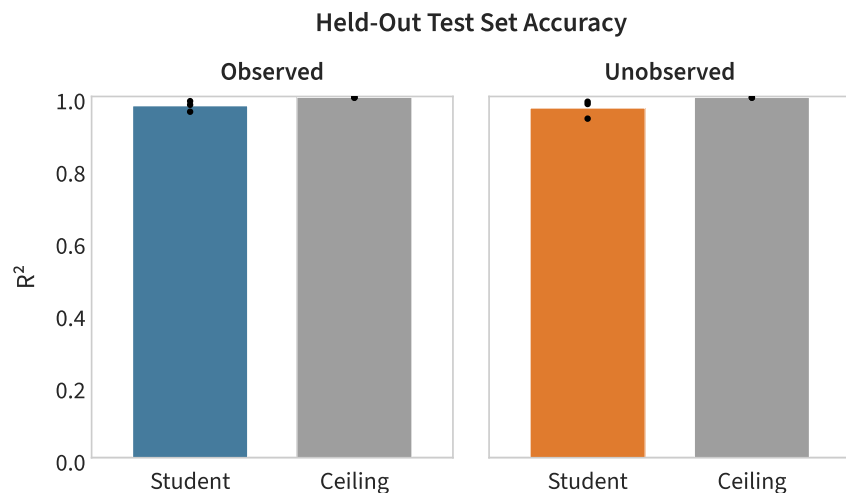
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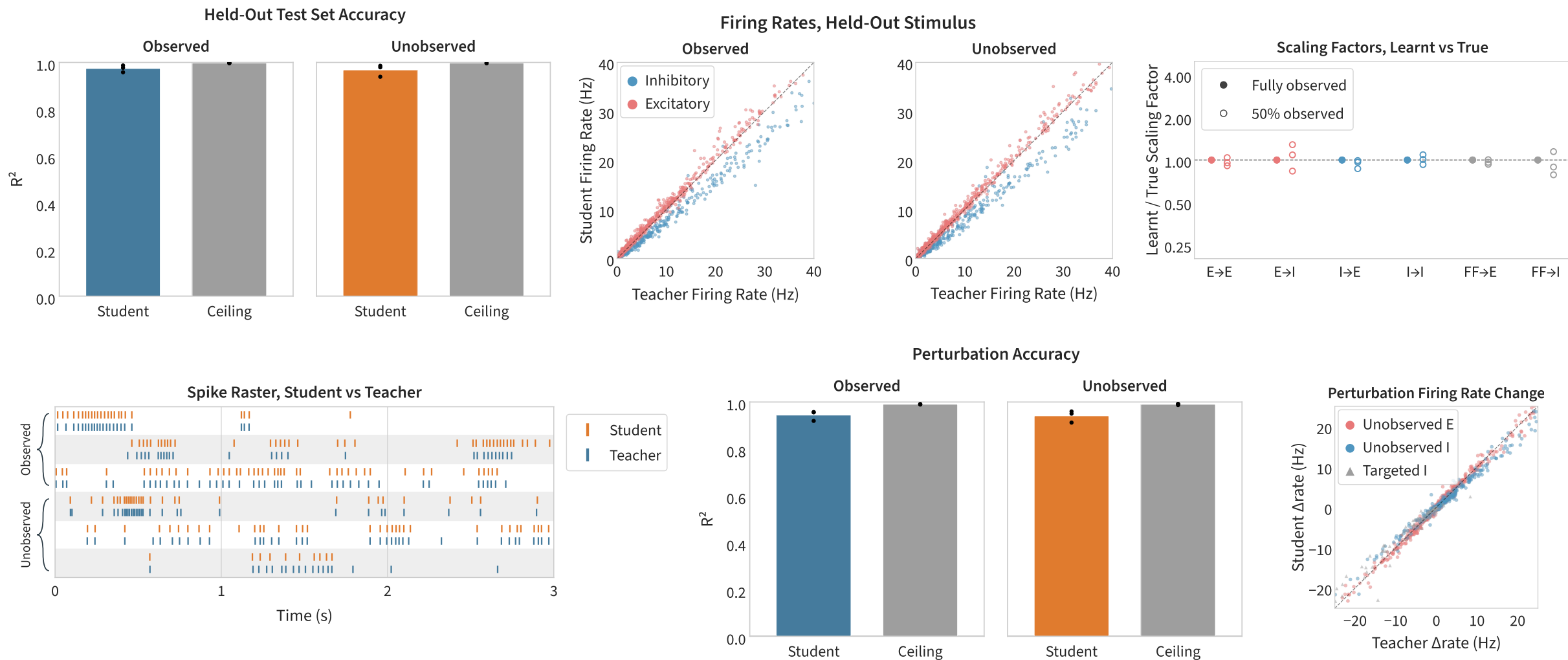
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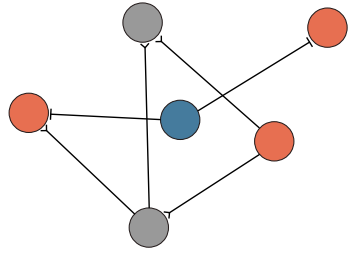
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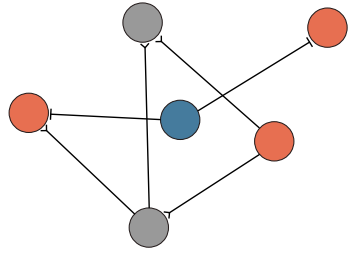
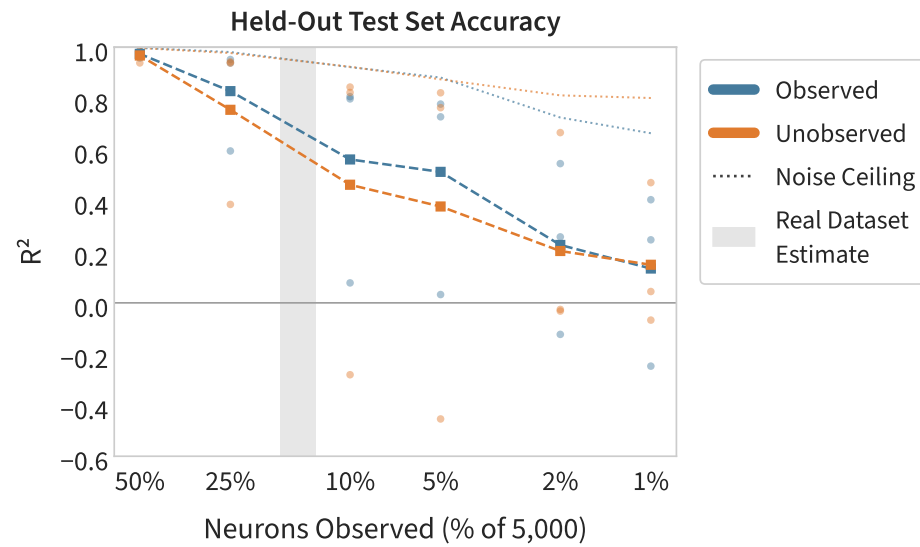
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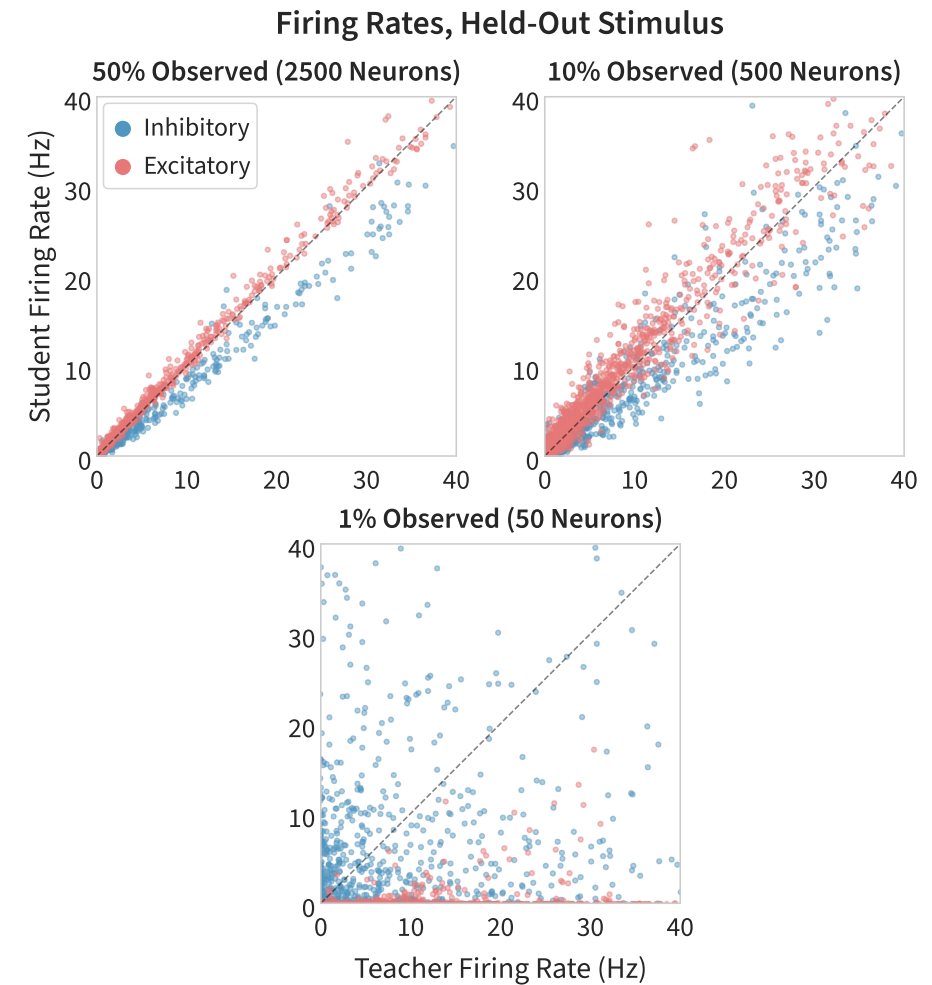
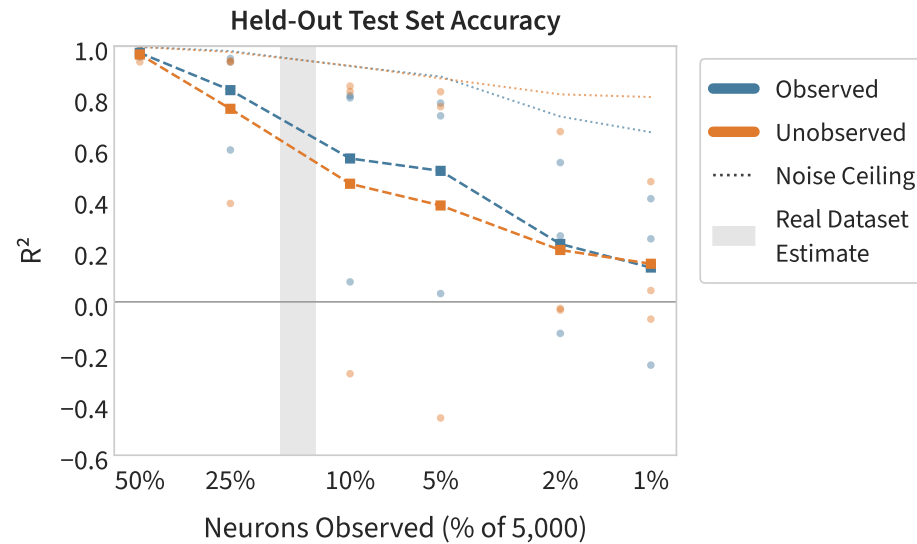
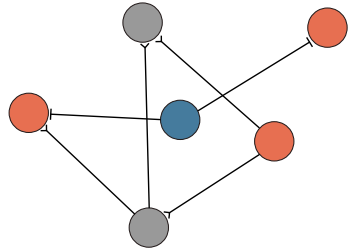
Robust to a Fraction of Neurons Recorded

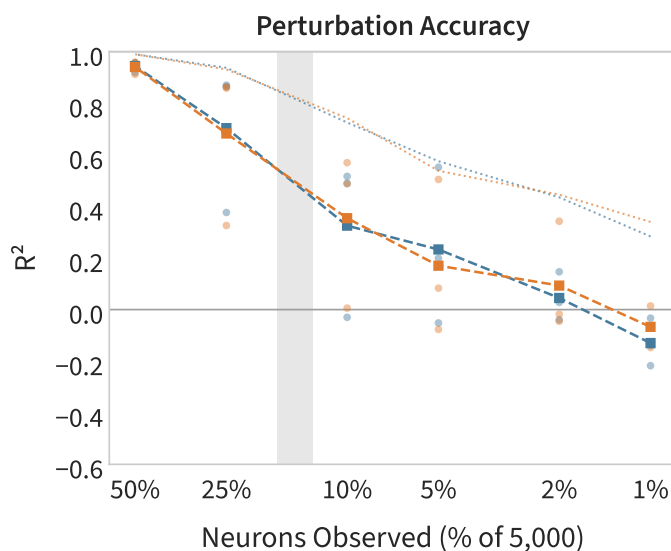


Robust to a Fraction of Neurons Recorded

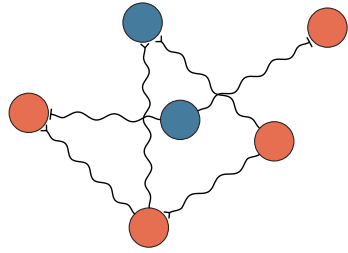


Robust to a Fraction of Neurons Recorded

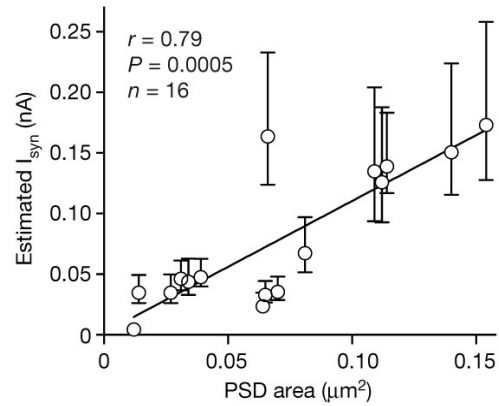
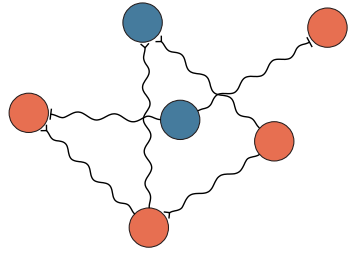




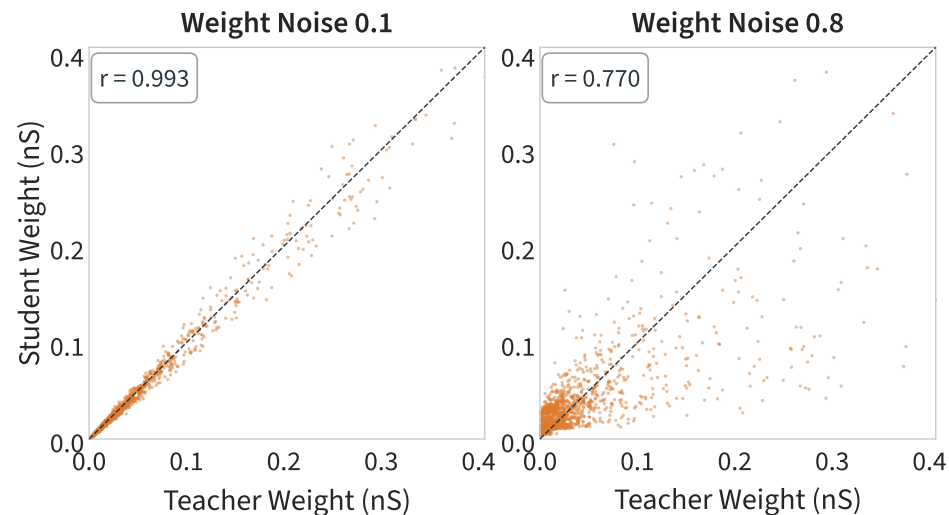
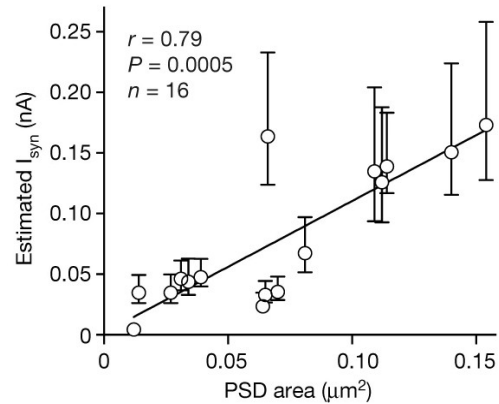
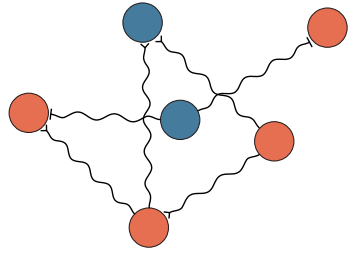
Robust to Degrading the Correlation Between Synapse Volume and Weight



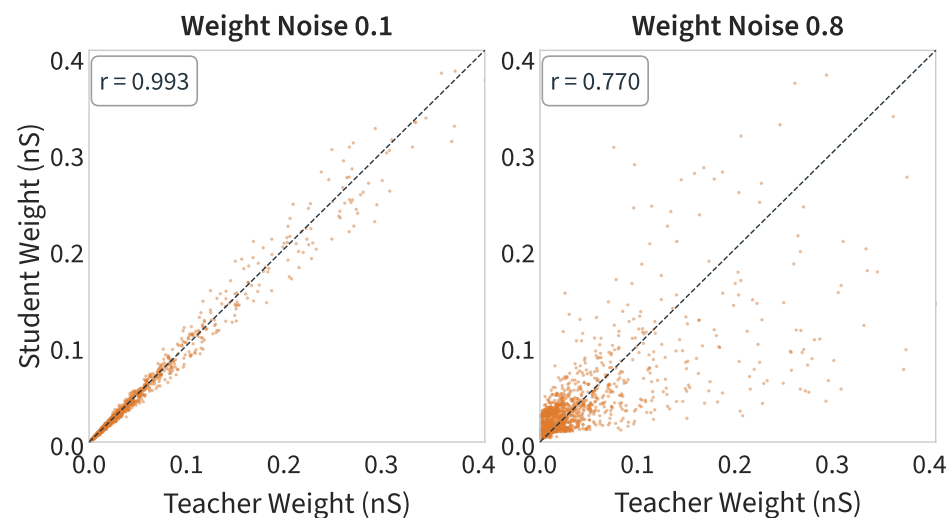
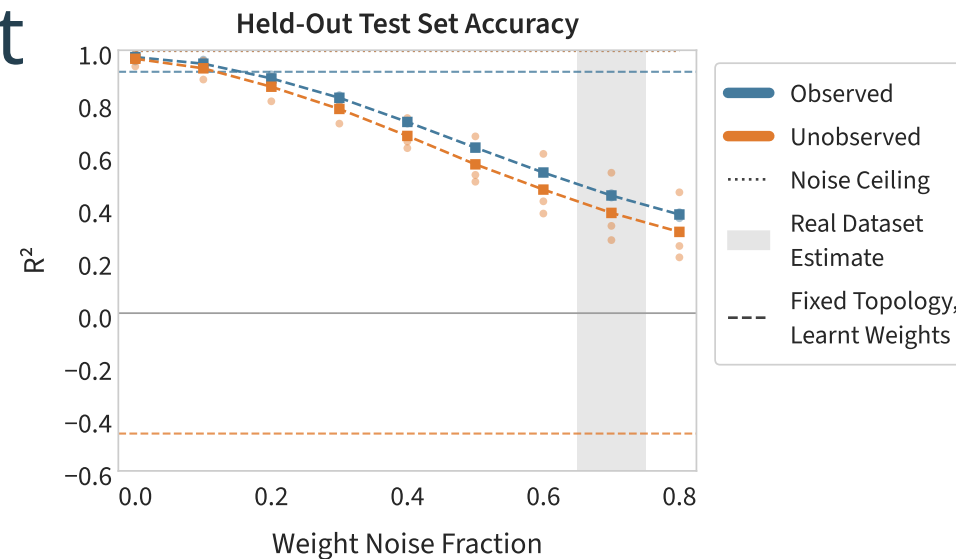
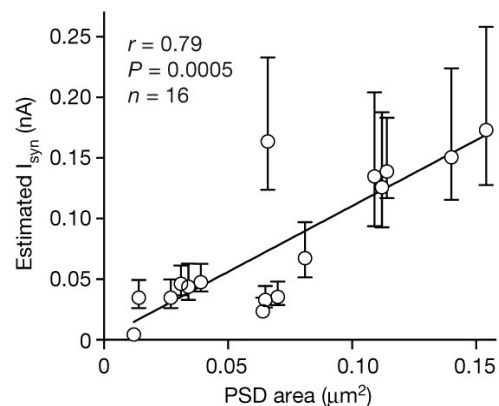
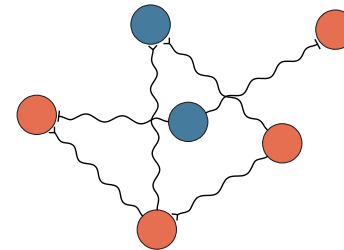
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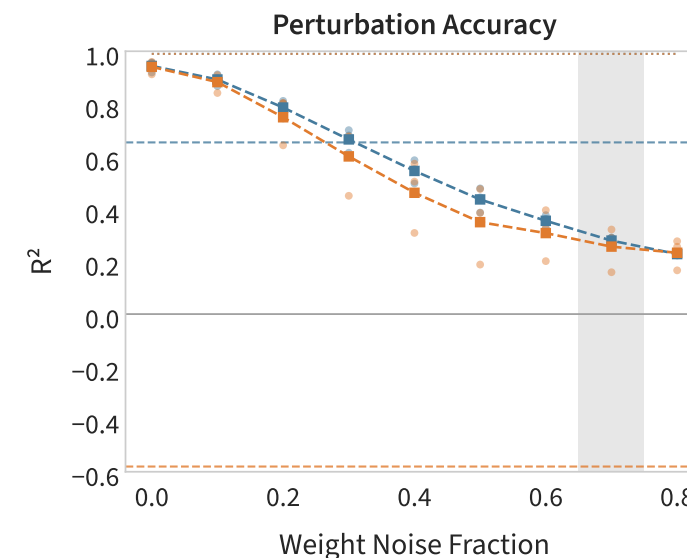
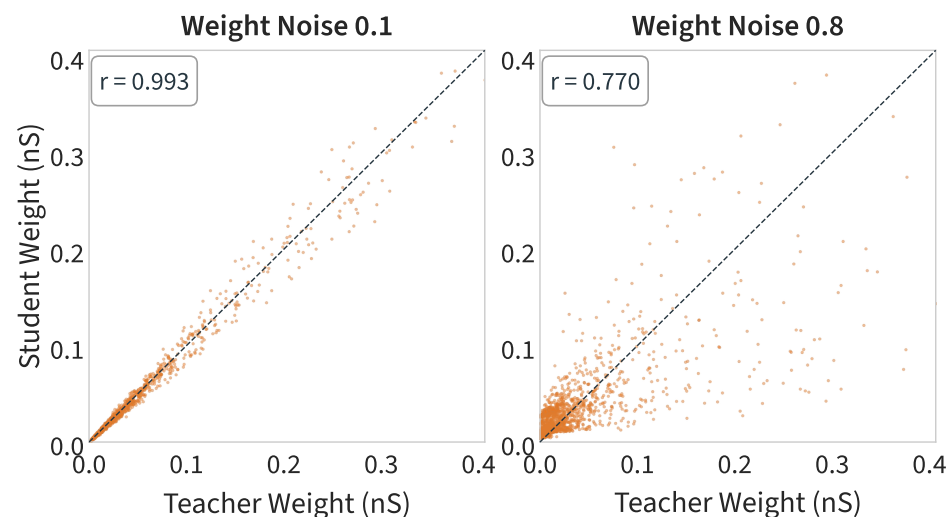
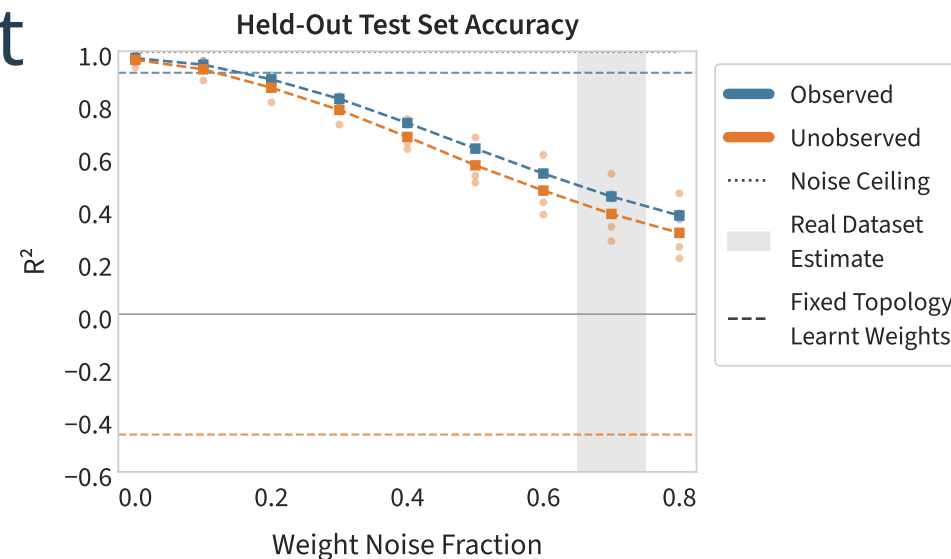
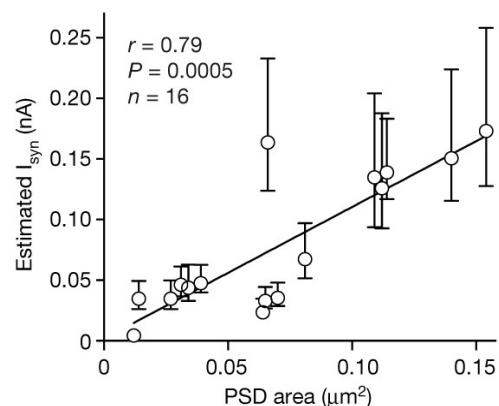
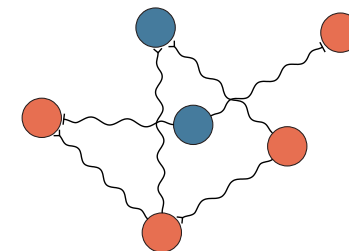
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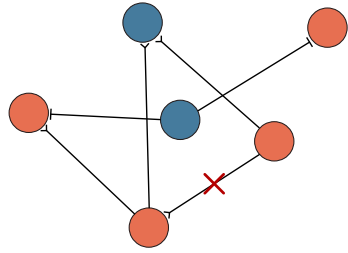
Robust to Degrading the Correlation Between Synapse Volume and Weight



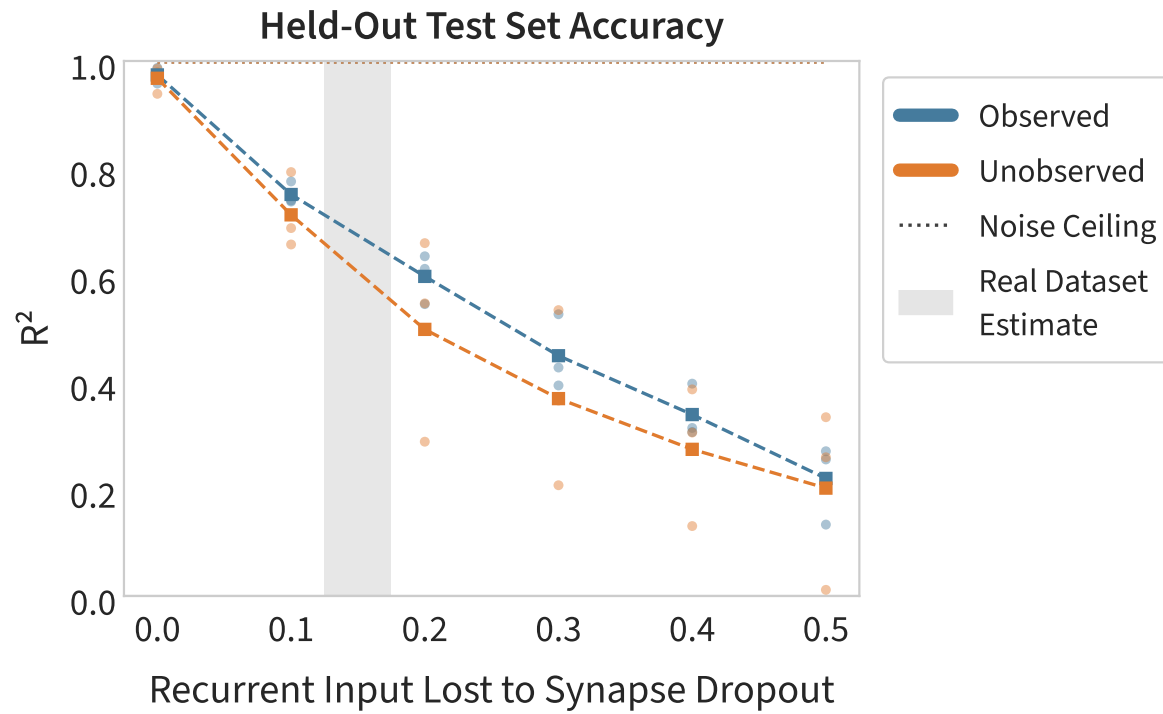
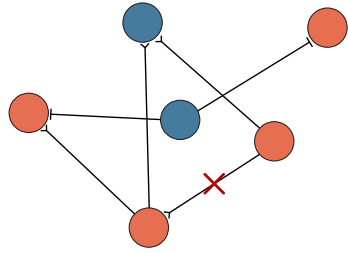
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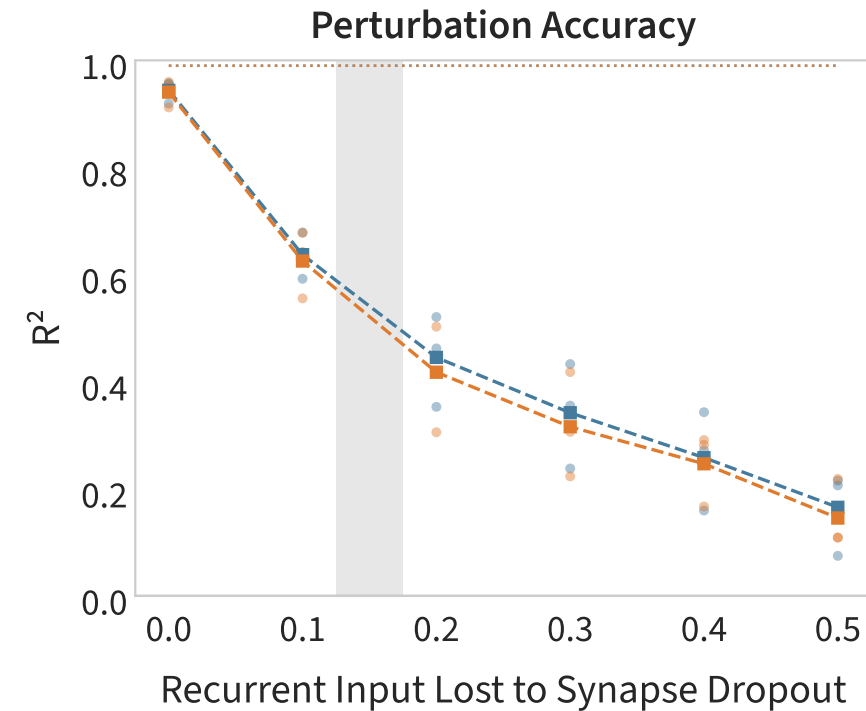
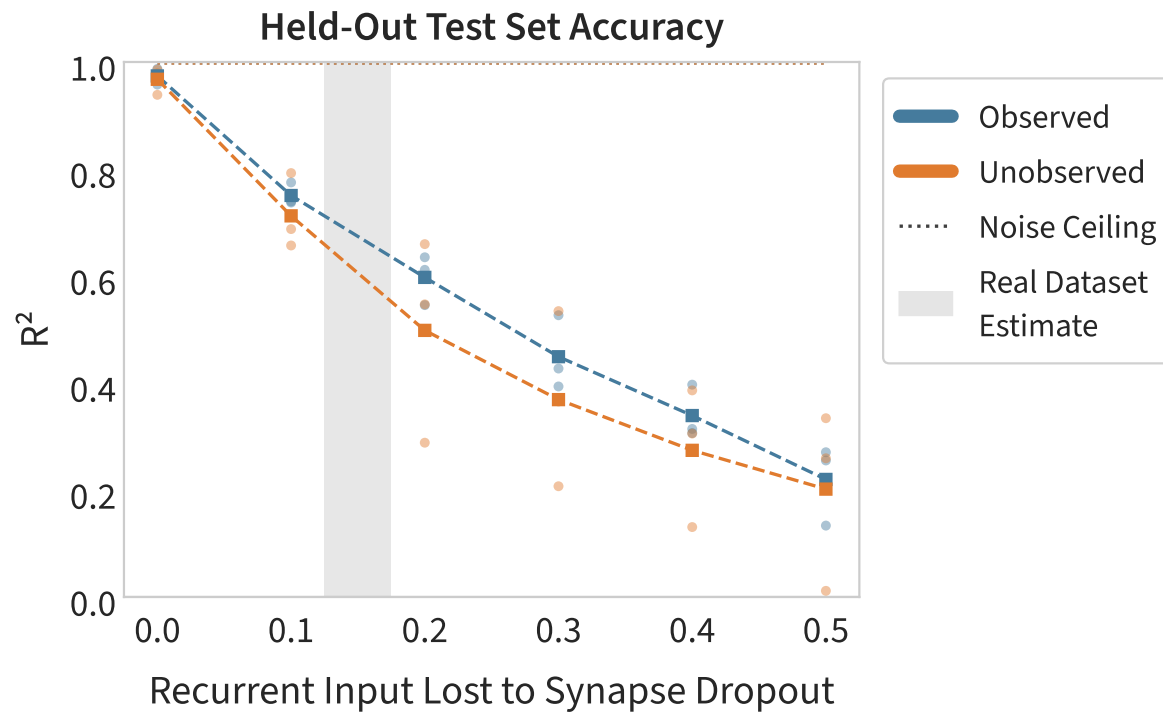
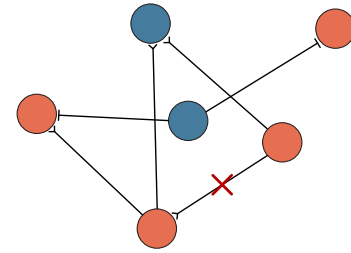
Robust to Connectome Reconstruction Errors



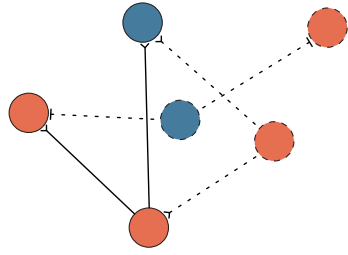
Robust to Connectome Reconstruction Errors



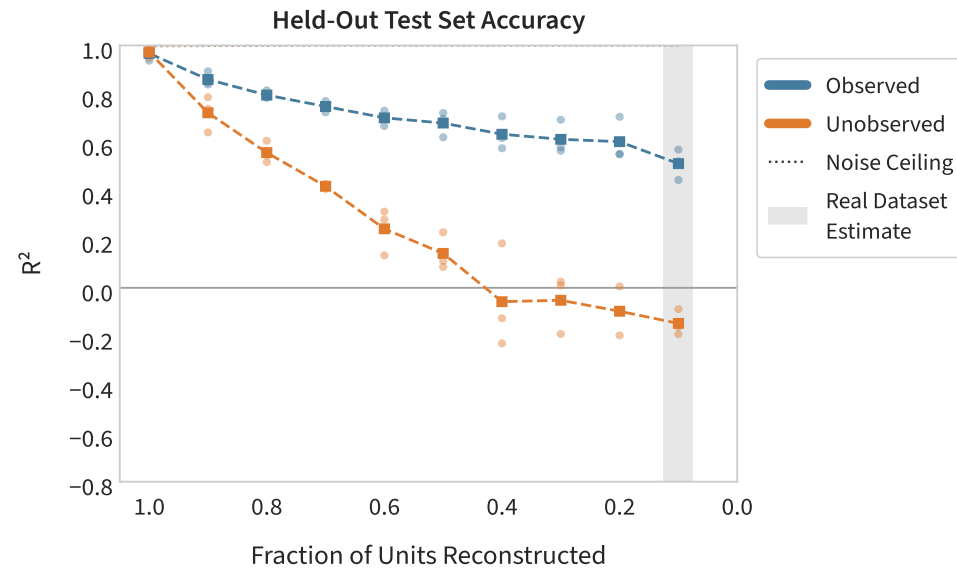
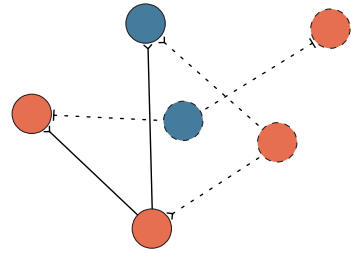
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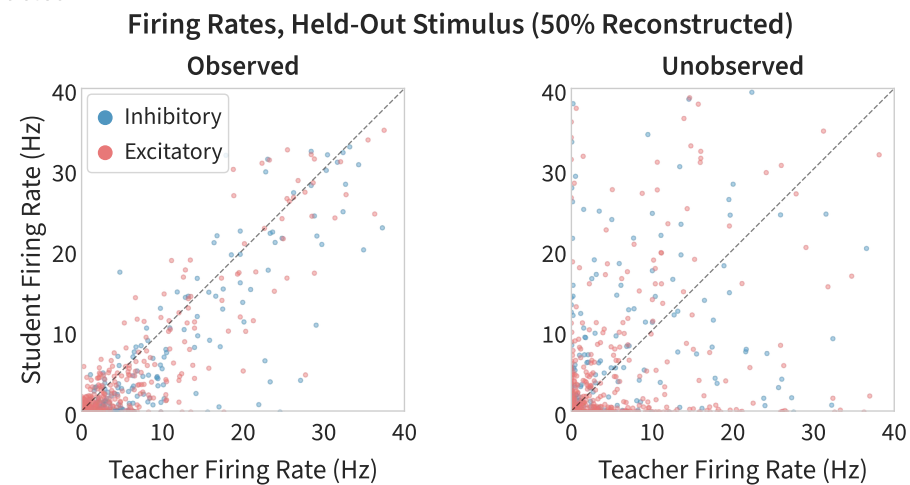
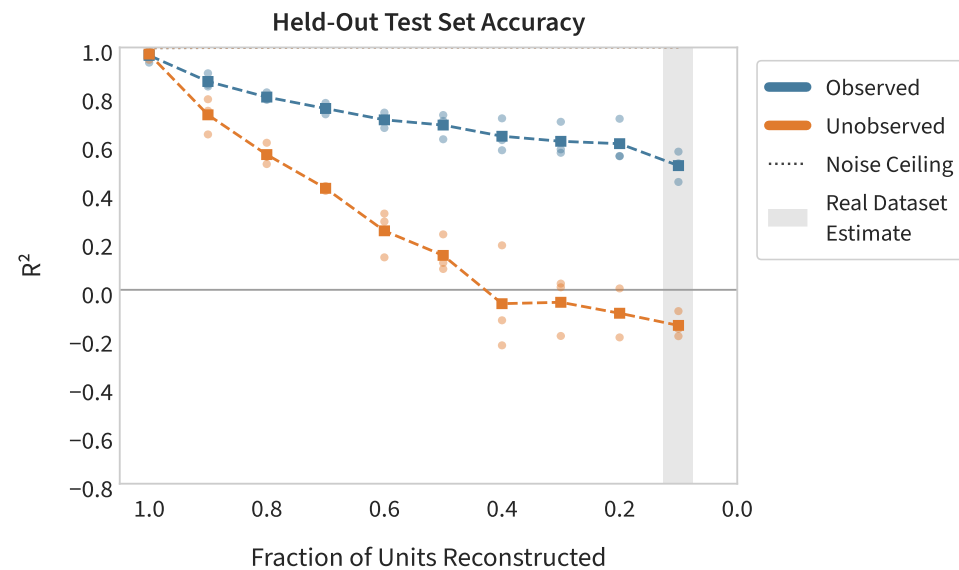
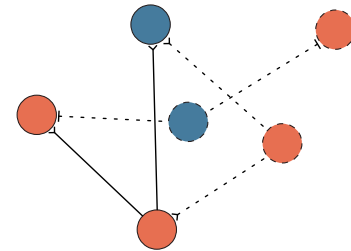
Performance Drops Rapidly with Connectome Partial Reconstruction



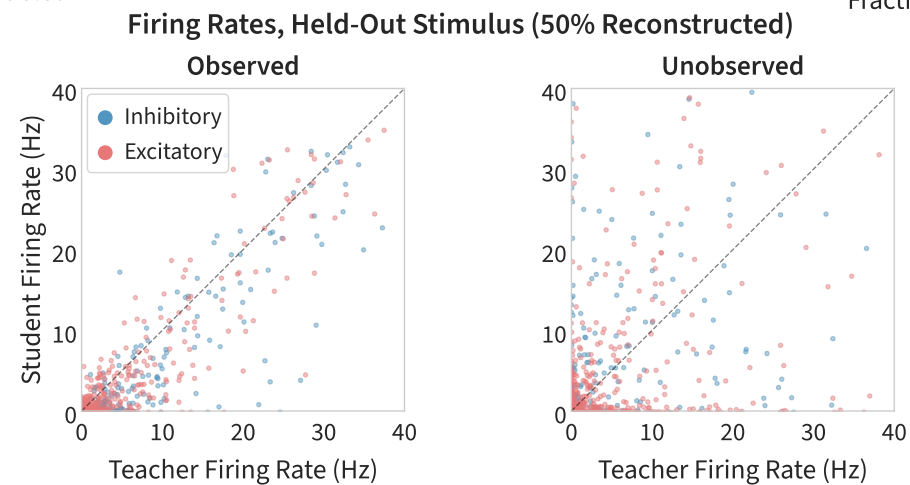
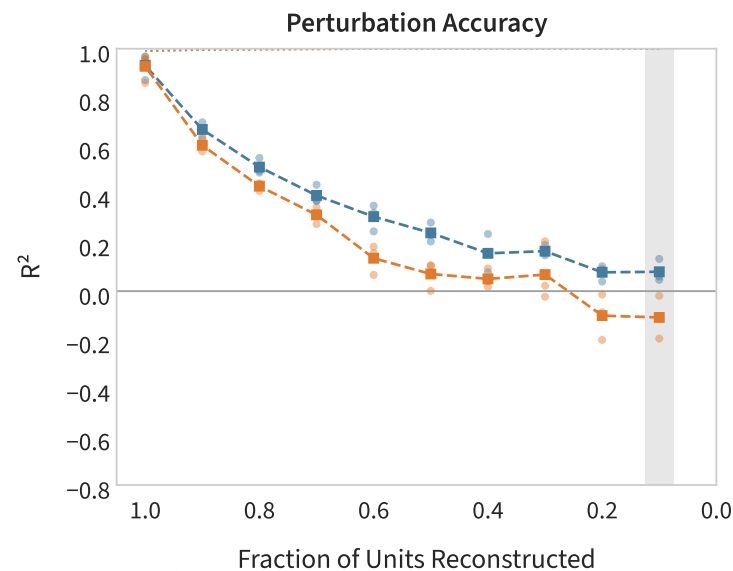
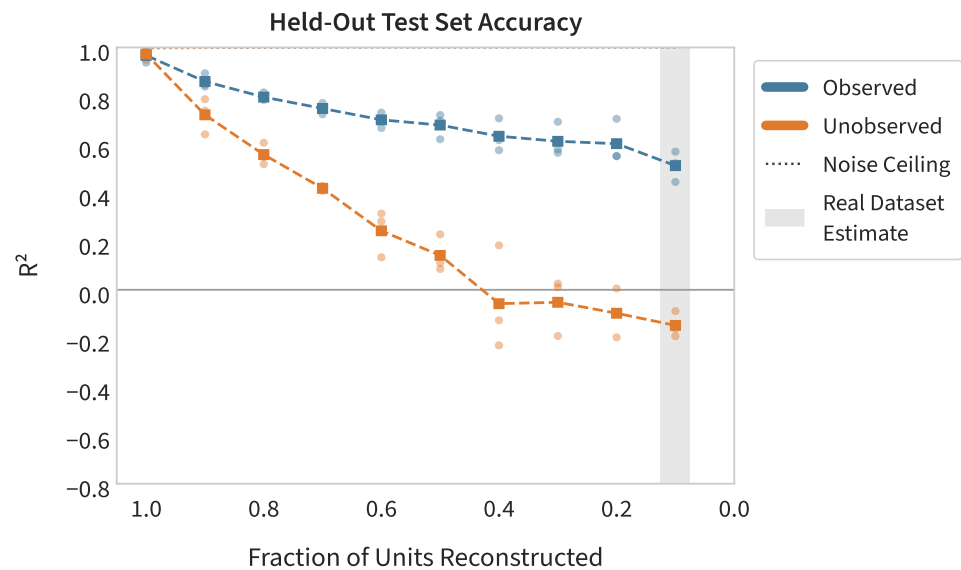
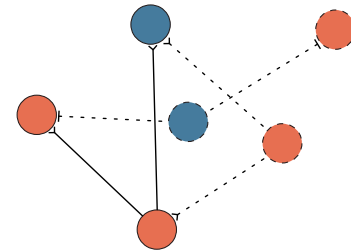
Performance Drops Rapidly with Connectome Partial Reconstruction



Performance Drops Rapidly with Connectome Partial Reconstruction



Performance Drops Rapidly with Connectome Partial Reconstruction



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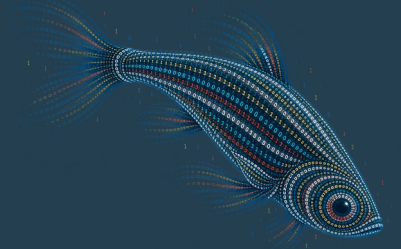
- Mechanistic circuit fits are possible if all the local circuit's inputs have been densely connectomically reconstructed
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Conclusion

- Mechanistic circuit fits are possible if all the local circuit's inputs have been densely connectomically reconstructed
- With only partial reconstruction, we are still in a highly degenerate regime
- This has encouraged us to focus on more densely reconstructing the circuit
- For Zebrafish larvae, with $\sim 1,100$ place cells, dense reconstruction is tractable
- We are exploring using Bayesian active learning to determine which inputs to selectively reconstruct to constrain our model the most

Thank You

github.com/
rory-bedford/teacher-student



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Tomas Gancarcik
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Mohammadi
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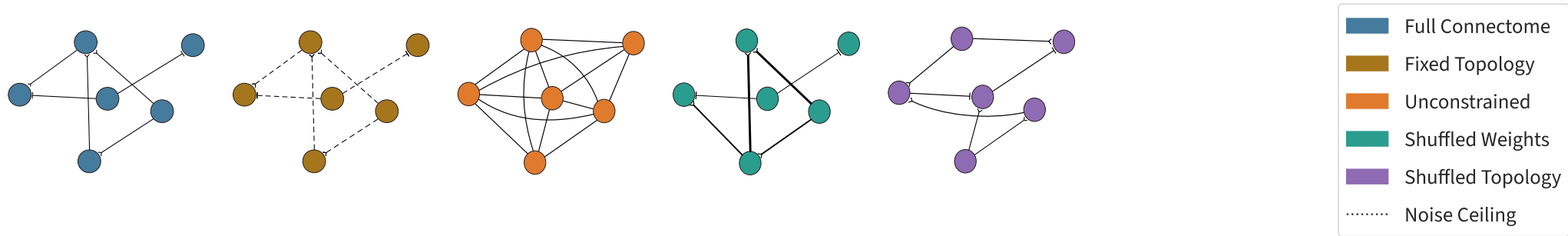
Chuyu Yang
Jennifer Li
Drew Robson



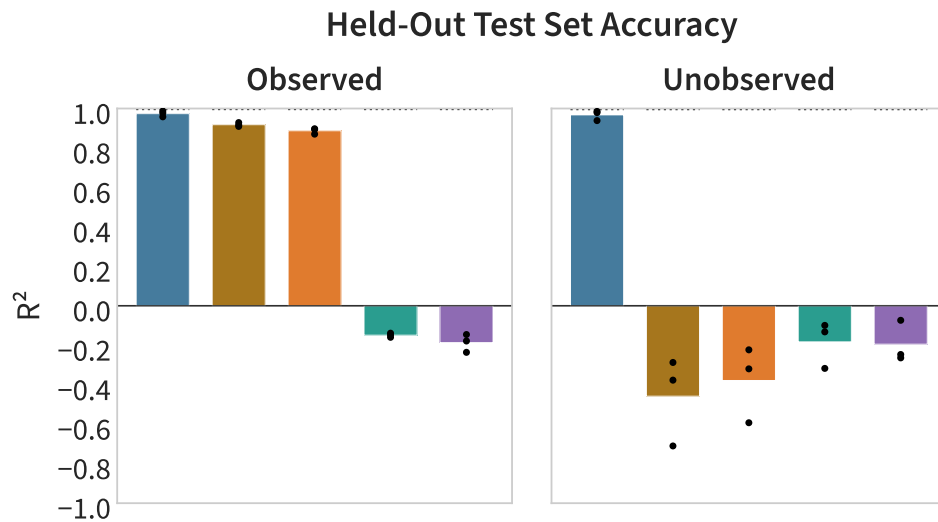
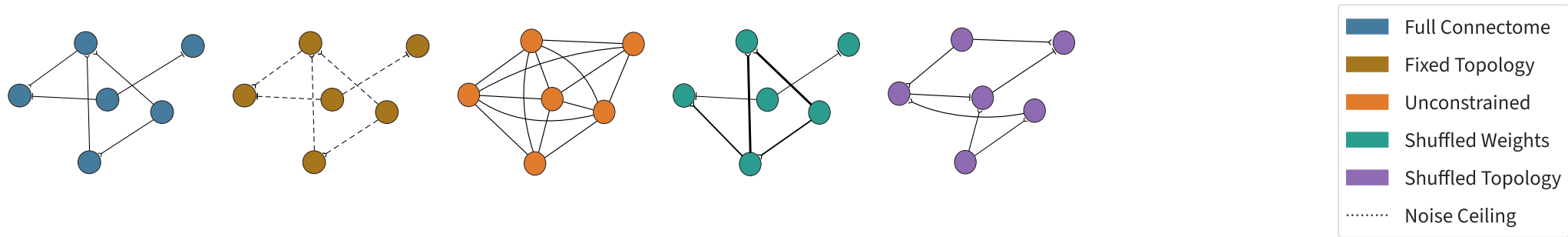
FMI IT Support
FMI Administrative Support

Connectome Beats Controls

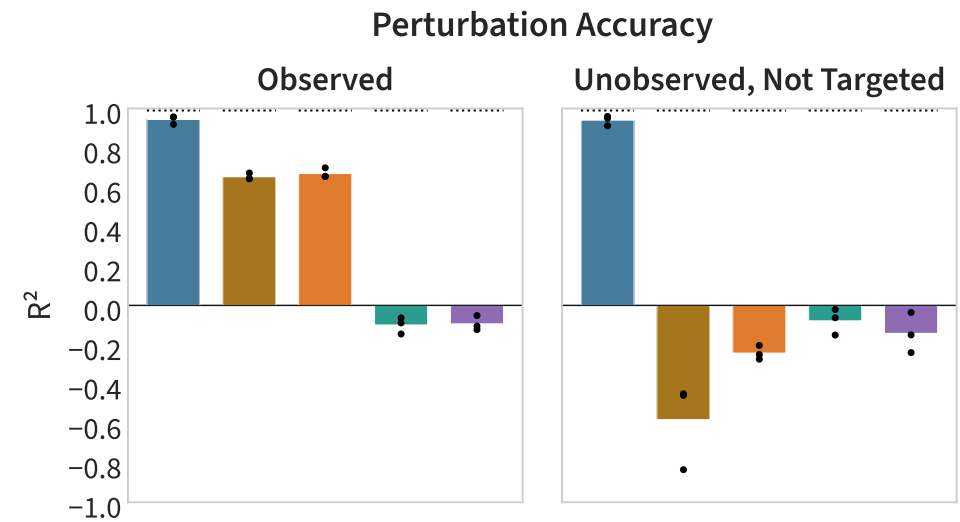
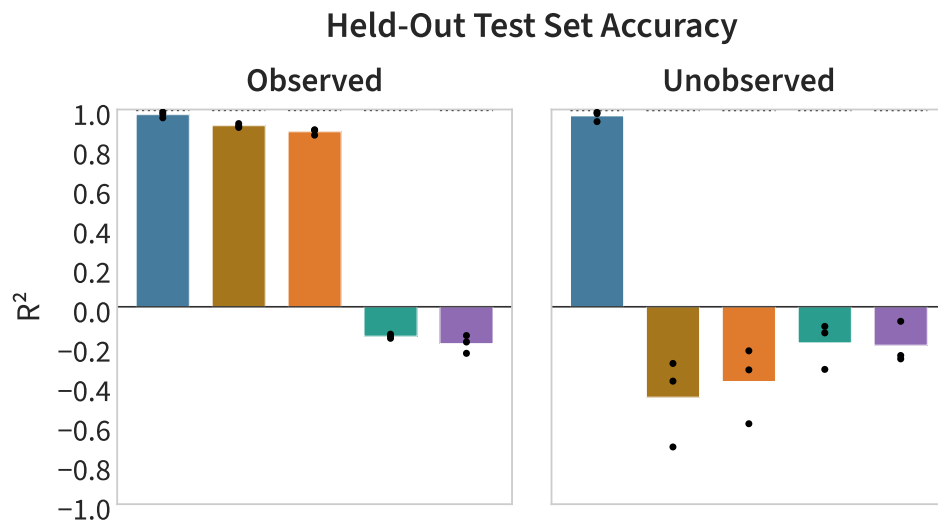
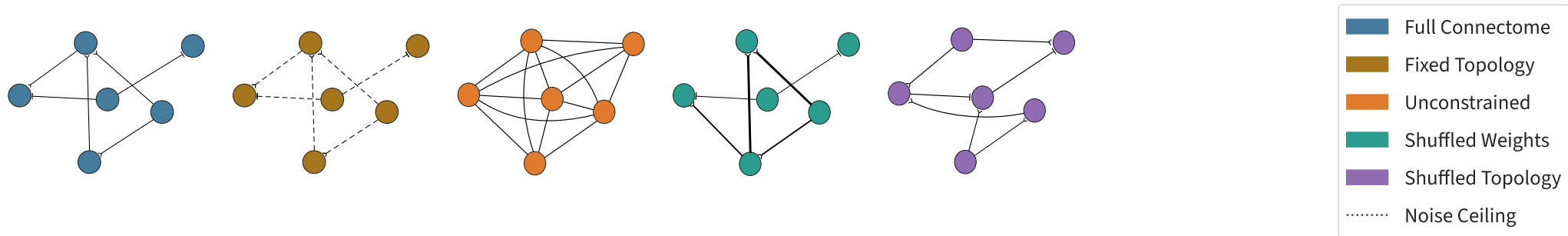
Connectome Beats Controls



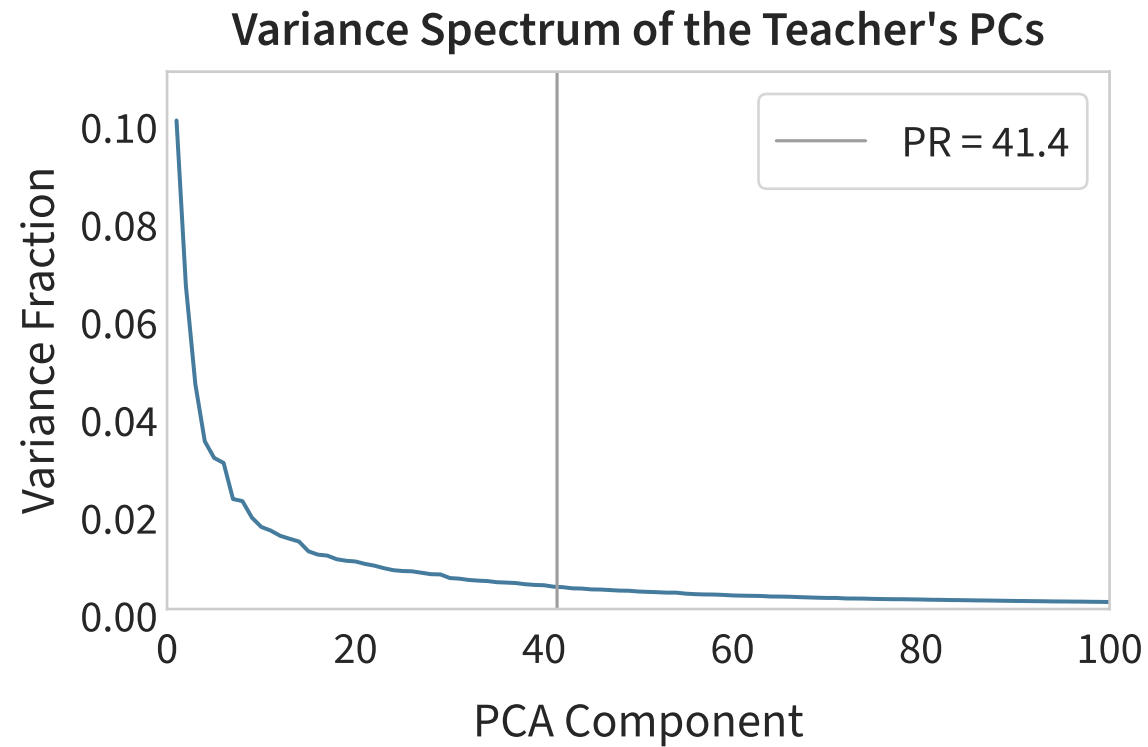
Connectome Beats Controls



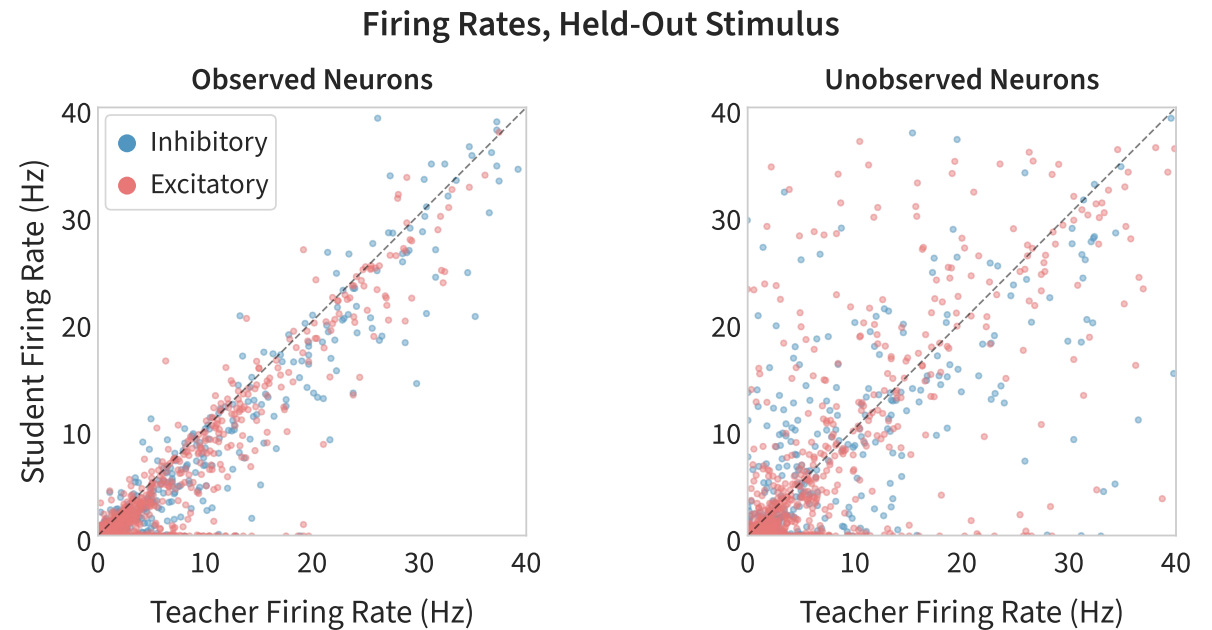
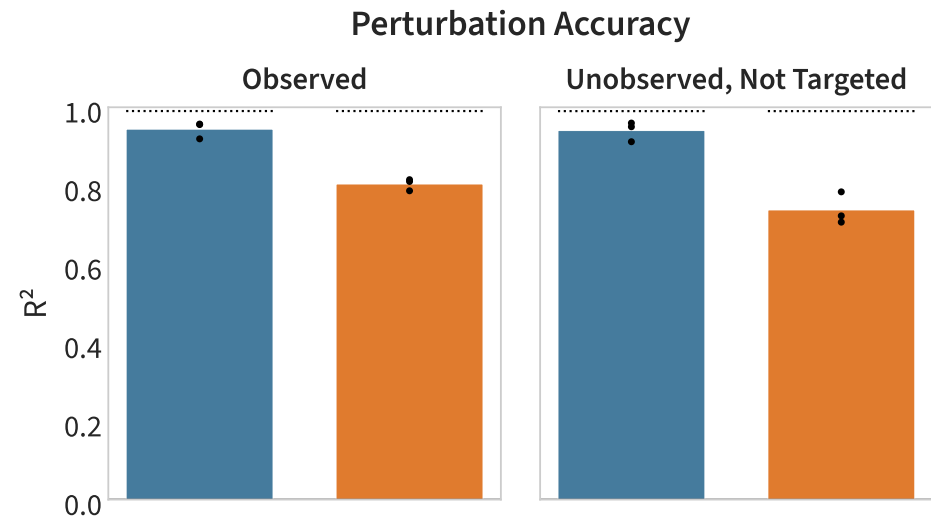
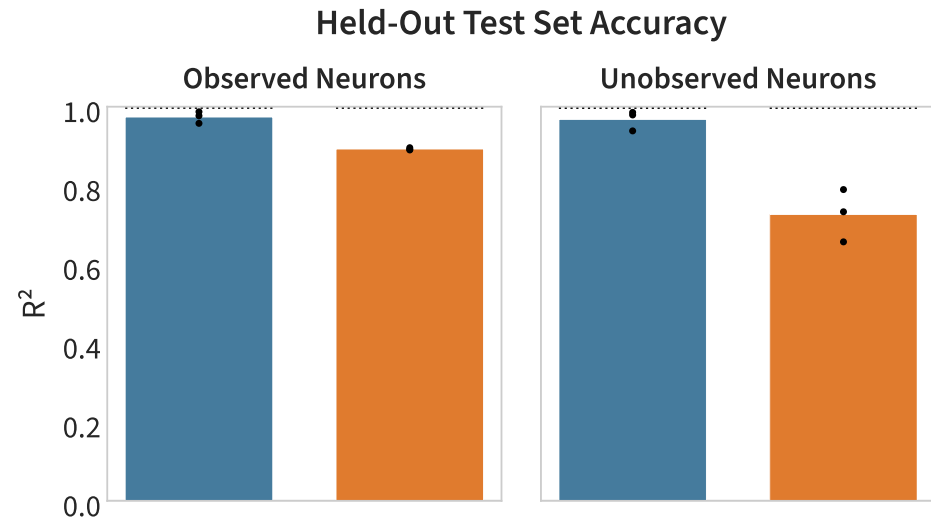
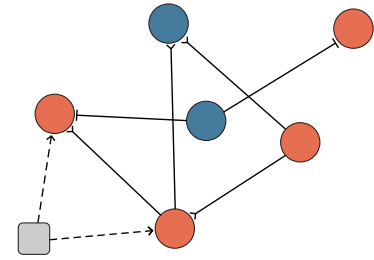
Connectome Beats Controls



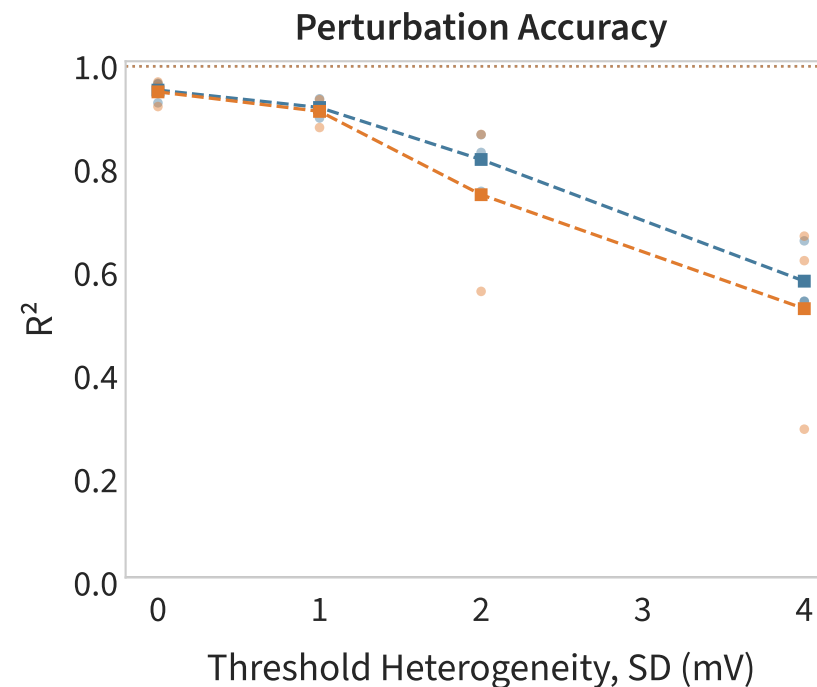
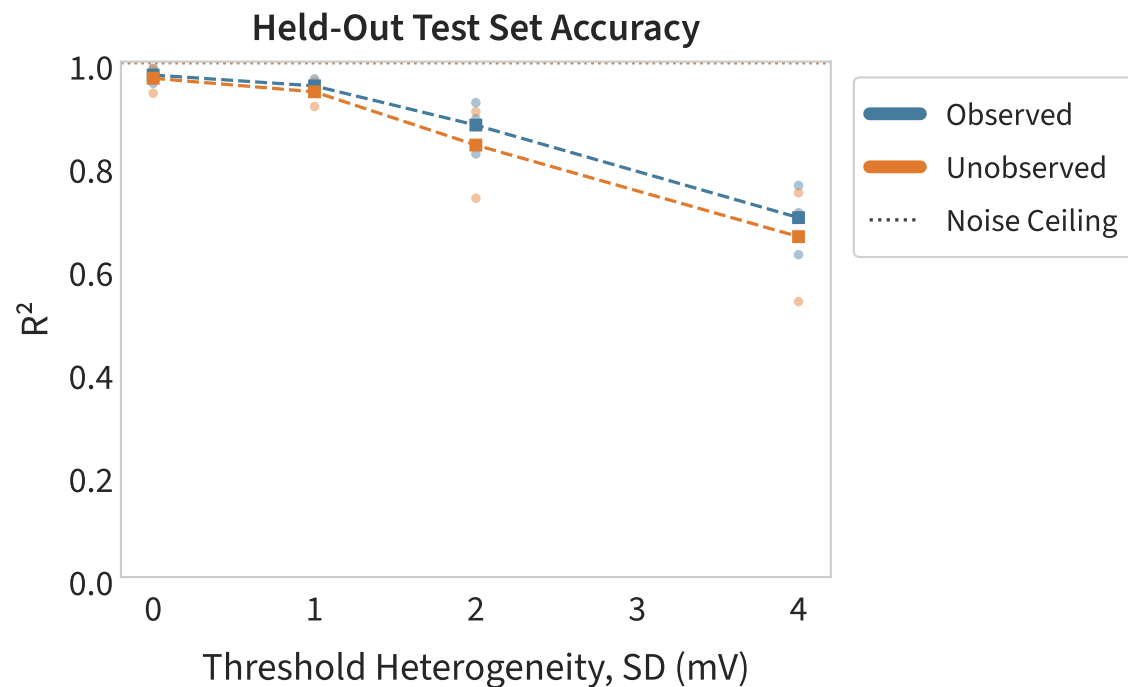
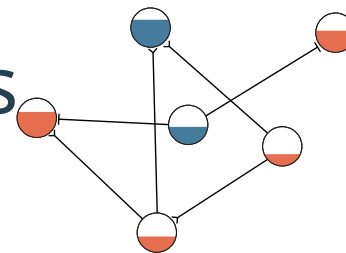
Teacher Network Dimensionality



Learning Input Connections with Recurrence Fixed



Model Mismatch – Heterogeneous Student Thresholds



Non-Reconstructed Neurons

